

Automated Risk of Bias Assessment in Systematic Reviews Using Large Language Models: A Multi-Model Single-Agent Benchmark on 227 Randomised Controlled Trials

Luiz Felipe Guain Teixeira^{1,2,3} Cassiano Pereira de Barros⁴
Renato Vicente^{1,2,3}

¹TELUS Digital Research Hub

²Center for Artificial Intelligence and Machine Learning, University of São Paulo

³Institute of Mathematics, Statistics and Computer Science, University of São Paulo

⁴Paulista School of Medicine, Federal University of São Paulo

{luiz.felipe.teixeira, rvicente}@usp.br

Abstract

Background: Risk of bias (RoB) assessment is a mandatory but labour-intensive step in systematic review production. Large language models (LLMs) offer the potential to automate this task in a zero-shot setting, yet systematic benchmarks spanning multiple frontier models and inference strategies are lacking.

Methods: Eight state-of-the-art LLMs (Gemini 2.5 Pro, Gemini 2.5 Flash, GPT-5.4, Claude Haiku 4.5, DeepSeek v4-Flash, Grok 4.3, Qwen 3.7-Plus, and GLM-5)

were evaluated on zero-shot, single-agent RoB 1.0 assessment of 227 randomised controlled trials drawn from 127 Cochrane systematic reviews published in 2024. Models were accessed via the OpenRouter API in June 2026 using a fixed prompt and provider-default generation parameters (exact OpenRouter identifier strings in Table 2). Ground-truth annotations were taken from the published Cochrane consensus judgements across seven methodological domains (1,589 cells total). Performance was assessed using accuracy, Macro-F1, and Gwet's AC1, with study-clustered bootstrap 95% confidence intervals. Two models (Gemini 2.5 Flash and DeepSeek v4-Flash) were additionally evaluated under a single-call inference mode for comparison with the per-domain strategy.

Results: Overall accuracy ranged from 57.6% to 62.5% across models, a spread smaller than the width of individual 95% confidence intervals; Macro-F1 confidence intervals overlapped across all eight models. AC1 provided the clearest separation: Claude Haiku 4.5 ($AC1 = 0.485$; 95% CI $[0.443, 0.526]$) had non-overlapping CIs relative to Gemini 2.5 Pro ($[0.344, 0.420]$) and GPT-5.4 ($[0.349, 0.426]$), though this comparison is unadjusted and exploratory. Unclear Risk was the most consistently under-detected class (recall < 0.20 in three of seven domains across most models). Consensus failures (cells where no model was correct) accounted for only 14.2% of the benchmark, confirming substantial non-redundancy among models; the remaining 85.8% of cells had at least one correct model. The comparison of inference modes yielded small, largely non-significant differences: ten of twelve Δ metrics had confidence intervals spanning zero, and D7 (Other Bias)—the domain hypothesised to benefit most from joint assessment—showed negligible change under either mode.

Conclusions: Frontier LLMs can perform zero-shot RoB 1.0 assessment at accuracy levels broadly consistent with estimates of human inter-rater variability for this task [1], though direct comparison on this specific corpus was not performed; no single model dominates across all metrics, and no consistent advantage was found for single-call over per-domain inference.

46 The primary practical distinction between inference modes is computational:
47 single-call reduces inference calls by $7\times$ without a reliable accuracy trade-off.
48 All analyses are exploratory; results should be interpreted as directional
49 benchmarks rather than definitive rankings.
50 **Keywords:** risk of bias; systematic review; large language models; natural
51 language processing; Cochrane; evidence synthesis; automation

1 Introduction

Systematic reviews and meta-analyses occupy the apex of the evidence hierarchy in evidence-based medicine [2]. Their validity depends critically on the quality of the primary studies included: a meta-analysis that aggregates trials with uncontrolled confounding or selective outcome reporting inherits—and can amplify—those methodological flaws. The Cochrane Risk of Bias 1.0 tool (RoB 1.0) [3] was developed precisely to render this quality appraisal explicit and reproducible. By requiring reviewers to judge each included trial across seven pre-specified methodological domains, RoB 1.0 transforms a subjective quality impression into a structured, auditable record.

Despite this formalisation, RoB assessment remains a major bottleneck in systematic review production. Each trial must be read by two independent reviewers, who compare judgements and resolve disagreements through discussion or adjudication. For a review covering dozens of trials, this process routinely demands tens to hundreds of person-hours. Inter-rater reliability, even among trained reviewers, is moderate at best [4, 5]—a consequence of genuine ambiguity in reporting, domain-specific expertise requirements, and the epistemic character of the *Unclear Risk* category, which denotes insufficient information rather than a graded risk level. These limitations have motivated sustained interest in automating, or at least augmenting, the RoB assessment workflow.

Early automation efforts relied on rule-based systems and supervised machine learning trained on sentence-level text fragments [6]. Such approaches demonstrated that natural language processing could identify methodological descriptions in trial reports with above-chance accuracy, but they were constrained by limited contextual understanding, sensitivity to reporting style, and the need for substantial labelled training data for each domain.

Large language models (LLMs) trained on diverse corpora with instruction-following capabilities represent a qualitatively different paradigm [7, 8]. Without task-specific fine-tuning, general-purpose LLMs can be prompted to read

81 a full trial report, locate domain-relevant passages, and produce structured
82 judgements with free-text justifications—closely mimicking the expert reviewer
83 workflow. A prompt template can address all seven RoB 1.0 domains either
84 sequentially—one domain per inference call—or simultaneously in a single call;
85 whether this choice affects assessment quality is an open empirical question.
86 As frontier LLMs have grown more capable and their inference costs have fallen,
87 evaluating their effectiveness for RoB assessment has become both feasible
88 and practically important. A recent preregistered comparative study applied
89 four contemporary LLMs to 100 RCTs from Cochrane reviews under the RoB 1.0
90 instrument, reporting interobserver κ of 0.27–0.39 and concluding that none
91 reached the reliability threshold required for autonomous use [9]. Whether this
92 shortfall is uniform across the RoB 1.0 structure—or whether reliable assistance
93 is already viable for specific domains and intervention types while systematic fail-
94 ures persist elsewhere—has direct implications for how LLMs can be integrated
95 into real screening workflows.

96 The present study addresses this question with a systematic comparison
97 of eight state-of-the-art LLMs on zero-shot, single-agent RoB 1.0 assessment
98 applied to 227 randomised controlled trials drawn from 127 Cochrane systematic
99 reviews published in 2024. All models were evaluated on a common benchmark
100 under identical conditions: the same prompt, the same fully text-based study
101 representations produced by a standardised PDF-to-Markdown pipeline, and the
102 same ground-truth human annotations. Beyond aggregate accuracy and Macro-F1,
103 we report Cohen’s κ , Gwet’s AC1, and per-class F1—placing κ and AC1 side by side
104 to expose reliability estimates that diverge under class imbalance. Domain- and
105 intervention-stratified analyses identify where systematic biases arise and locate
106 the improvement opportunities most likely to yield practical gains for assisted
107 screening. Additionally, we compare two inference strategies—per-domain and
108 single-call—to assess whether querying all seven domains simultaneously affects
109 performance relative to sequential per-domain calls.

2 Methods

2.1 Dataset

The ground-truth dataset was constructed from Cochrane systematic reviews published in 2024. Of the 149 reviews published that year, 127 were accessible through our institutional subscription; from each accessible review, up to three included randomised controlled trials (RCTs) were selected on the basis of full-text PDF availability: all included RCTs for which a PDF was retrievable were included, up to a maximum of three per review, yielding a final corpus of **227 RCTs** drawn from 127 distinct Cochrane reviews.

Risk of bias annotations were obtained directly from the published Cochrane review reports and reflect the consensus judgements produced by the original review teams following standard Cochrane procedures (independent assessment by at least two trained reviewers, with disagreements resolved by discussion or arbitration). These annotations were not re-produced or independently adjudicated for the present study; they represent the published Cochrane consensus, which constitutes the reference standard used by the field. Although the revised Risk of Bias 2.0 tool (RoB 2; Sterne et al. 10) is now recommended for new systematic reviews, it reorganises the domain structure and changes the rating scale, rendering it incompatible with the existing RoB 1.0 corpus. RoB 1.0 remains the basis for the large majority of existing systematic review annotations — including all 127 reviews sampled here and most of the large labelled datasets used for training and evaluation of automated systems — so this study’s focus on RoB 1.0 is both corpus-driven and practically motivated. Each study was annotated using the Cochrane Risk of Bias 1.0 (RoB 1.0) tool [3], which covers seven methodological domains: D1 (random sequence generation), D2 (allocation concealment), D3 (blinding of participants and personnel), D4 (blinding of outcome assessment), D5 (incomplete outcome data), D6 (selective outcome reporting), and D7 (other sources of bias). Each domain received one of three judgements—*Low Risk*, *High*

138 *Risk, or Unclear Risk*—yielding 1,589 annotated cells (227×7).

139 Class prevalence varied substantially across domains (Table 1). Low Risk was
140 the largest class in six of seven domains, and the majority class ($>50\%$) in four
141 of them (D1, D5, D6, D7); only D3 (blinding of participants and personnel) had
142 High Risk as its largest class (45%), reflecting the predominance of unblinded
143 pharmacological trials in the corpus. At the skewed extreme, D1 had a 69%
144 Low Risk majority and only 5 High Risk cases (2%); D5 and D7 were similarly
145 imbalanced at 63% and 61% Low Risk, respectively. D4 (blinding of outcome
146 assessors) was the most balanced domain overall (44% Low Risk, 28% High Risk,
147 29% Unclear Risk), closely followed by D2, in which Low Risk and Unclear Risk
148 were nearly equally represented (50% vs. 46%).

Table 1: Ground-truth class distribution per RoB domain ($n = 227$ studies per domain). LR = Low Risk; HR = High Risk; UR = Unclear Risk.

Domain	LR	HR	UR	Majority class	Majority (%)
D1	157	5	65	Low Risk	69.2
D2	113	10	104	Low Risk	49.8
D3	74	103	50	High Risk	45.4
D4	99	63	65	Low Risk	43.6
D5	143	39	45	Low Risk	63.0
D6	125	25	77	Low Risk	55.1
D7	139	31	57	Low Risk	61.2
Total	850 (53.5%)	276 (17.4%)	463 (29.1%)	Low Risk	53.5

149 **Intervention type.** Each study was assigned a single intervention category:
150 drug/medication ($n = 104$), non-pharmacological intervention ($n = 75$), proce-
151 dure ($n = 30$), or medical device ($n = 18$).

152 **Clinical domain.** Studies were tagged with one or more of 24 Cochrane Re-
153 view Group categories (multi-label; 120 of 227 studies belonged to more than
154 one group). The most frequent categories were Child Health ($n = 72$), Heart &
155 Circulation ($n = 45$), and Kidney Disease ($n = 31$); the least frequent were Skin
156 Disorders ($n = 3$), Effective Practice & Health Systems ($n = 4$), Rheumatology
157 ($n = 5$), and Wounds ($n = 5$). For stratified analyses, each multi-domain study

158 was included in every applicable group.

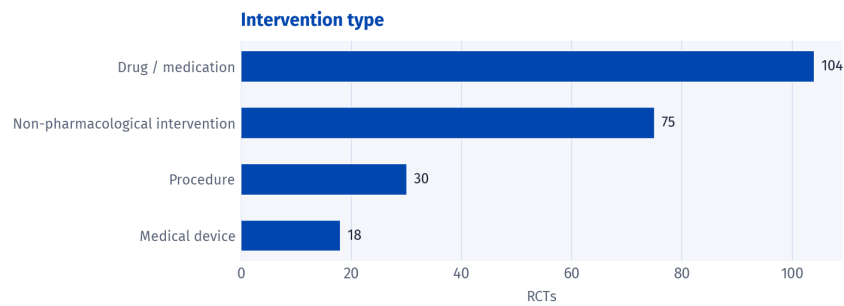


Figure 1: Distribution of the 227 RCTs by intervention type.

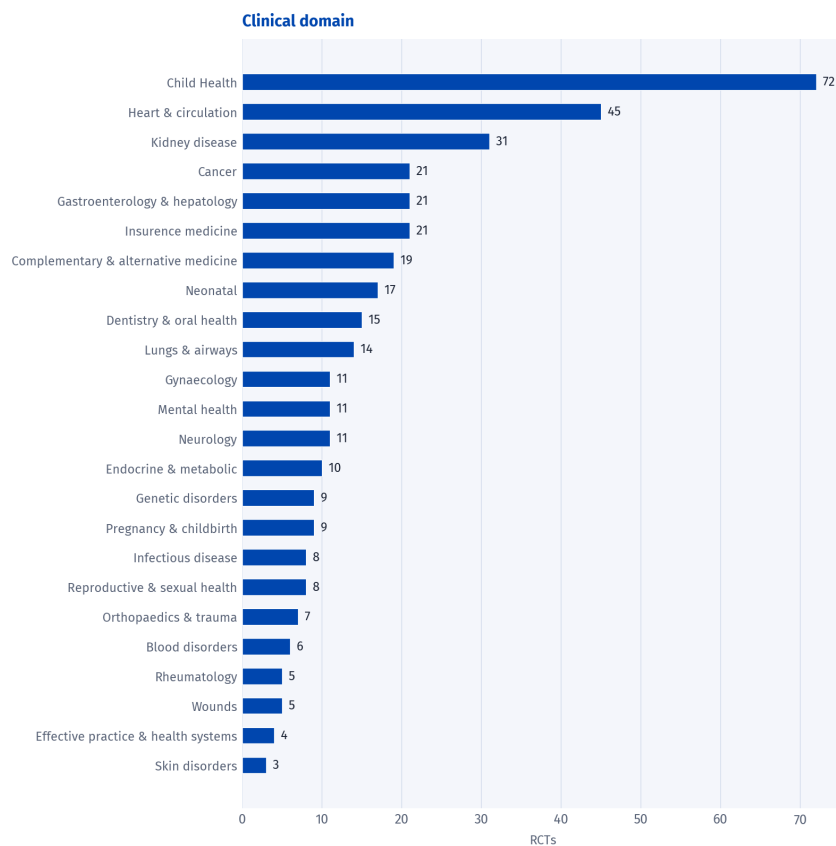


Figure 2: Distribution of the 227 RCTs by clinical domain (Cochrane Review Group categories, multi-label; total exceeds 227).

159 2.2 Document Preparation Pipeline

160 Full-text study reports (PDF) were processed through a two-stage pipeline that
161 converts each article into a standardised plain-text representation before it
162 is submitted to any model. Converting to a common format ensures that all

163 eight models receive identical textual input, so that observed differences in
164 performance reflect model behaviour rather than differences in how each model
165 internally parses or prioritises content from a raw PDF.

166 **Stage 1 — PDF-to-Markdown conversion.** Each PDF was converted to struc-
167 tured Markdown using MinerU [11], which preserves headings, tables, and inline
168 text whilst replacing embedded figures with placeholder tokens of the form
169 [Figure k].

170 **Stage 2 — Figure description.** Each figure placeholder was resolved by Gem-
171 ini 2.5 Flash (google/gemini-2.5-flash, accessed via OpenRouter in May–June 2026),
172 a vision-language model (VLM) that received the corresponding page image
173 together with surrounding article context and generated a structured textual
174 description following a fixed eight-field schema (figure type, axes and units, leg-
175 end and groups, visual summary, plotted patterns, and unreadable or uncertain
176 areas; full prompt reproduced in Supporting Information S10). The description
177 was then inserted in place of the placeholder. After Stage 2, the document was
178 a self-contained Markdown file in which all information—including any figures
179 present in the source document—was represented as plain text. The result is
180 a self-contained Markdown file in which the full article content—text, tables,
181 and figures—is represented uniformly as plain text, forming the common input
182 submitted to all eight models.

183 **2.3 Risk of Bias Assessment**

184 All models were evaluated in a zero-shot, single-agent setting: each processed
185 Markdown document was submitted to the model with a fixed assessment prompt
186 (reproduced in full in Supporting Information S1), which specified the domain
187 criteria, the three admissible judgements (Low Risk, High Risk, Unclear Risk), and
188 the required JSON output format. No few-shot examples were provided, and the
189 prompt was used without modification across all models.

190 **Inference modes.** We evaluated two inference strategies that differ in how

the seven RoB domains are requested from the model. In the *per-domain* mode, seven sequential inference calls are made per study: each call supplies the full Markdown text together with a single domain identifier (e.g. Domain: D1), and the model returns a JSON object containing a judgement and justification for that domain only. In the *single-call* mode, a single inference call per study is made: the model receives the full Markdown text and is instructed to return a JSON array containing judgements and justifications for all seven domains simultaneously. All eight models were evaluated under the per-domain mode. Per-domain mode was adopted as the primary protocol because each RoB 1.0 domain represents an independent methodological criterion; separating calls ensures that each domain is assessed under a dedicated instruction context, without the model being required to simultaneously manage seven distinct evaluation tasks in a single generation pass. Gemini 2.5 Flash and DeepSeek v4-Flash were additionally evaluated under the single-call mode on the same 227 studies, enabling a controlled comparison of the two inference strategies whilst holding model, prompt content, and corpus constant. These two models were selected because their published list prices per token on OpenRouter were the lowest among the eight models evaluated, making a second full-corpus run economically practical. The results of this comparison are reported in Section 3.8.

2.4 Language Models

Eight frontier LLMs were evaluated (Table 2), accessed via the OpenRouter API in June 2026 using the identifier strings listed therein. No model-specific fine-tuning, prompt engineering beyond the fixed assessment prompt (Section 2.3), or retrieval augmentation was applied.

Evaluation protocol. All calls used provider-default generation parameters and each study-condition pair received a single inference draw with no output aggregation, constituting a hyperparameter-free single-draw evaluation protocol. This design reflects two constraints. First, decoding parameters such as

temperature interact with model architecture and training objective in model-family-specific ways; varying them across models would confound generation configuration with prompt design, rendering cross-model comparisons uninterpretable. Second, post-hoc reliability strategies—including self-consistency decoding [12], output ensembling, and majority voting over repeated draws—are optimisations orthogonal to native zero-shot capability and constitute a separate experimental question outside the scope of this benchmark.

Reproducibility and versioning. OpenRouter routes each identifier string to the provider-current checkpoint at inference time; underlying weights may be updated by the provider without a corresponding change to the public identifier. All results are therefore conditioned on the checkpoint states active during the June 2026 inference window and should not be assumed to generalise to subsequent revisions of the same identifier strings.

Table 2: Language models evaluated. All models accessed via the OpenRouter API in June 2026 using the identifier strings shown; zero-shot single-agent mode throughout; all 227 studies evaluated per model. Context length: maximum tokens accepted per call as reported by OpenRouter at access time. Release date: date the model became available on OpenRouter. Exact checkpoint versions are opaque at the provider level; the identifiers shown are the strings passed to the API and represent the model state at the time of inference.

Model	OpenRouter identifier	Context	Release
Gemini 2.5 Pro	google/gemini-2.5-pro	1M	Jun 2025
Gemini 2.5 Flash	google/gemini-2.5-flash	1M	Jun 2025
GPT-5.4	openai/gpt-5.4	1M	Mar 2026
Claude Haiku 4.5	anthropic/claude-haiku-4.5	200K	Oct 2025
DeepSeek v4-Flash	deepseek/deepseek-v4-flash	1M	Apr 2026
Grok 4.3	x-ai/grok-4.3	1M	Apr 2026
Qwen 3.7-Plus	qwen/qwen3.7-plus	1M	May 2026
GLM-5	z-ai/glm-5	203K	Feb 2026

2.5 Evaluation Metrics

Accuracy. The proportion of study-domain cells in which the model’s judgement matched the ground-truth label, equivalent to the observed agreement p_o used

in the chance-corrected coefficients below. Let $\mathbf{N} = (N_{ij})$ denote the $K \times K$ confusion matrix (here, $K = 3$), with rows indexed by predicted class i and columns indexed by ground-truth class j :

$$\text{Accuracy} = p_o = \frac{\sum_{k=1}^K N_{kk}}{N} = \frac{\sum_{k=1}^K N_{kk}}{\sum_{i=1}^K \sum_{j=1}^K N_{ij}}, \quad (1)$$

where N_{ij} is the number of cells with ground-truth class j predicted as class i , and $N = \sum_{i=1}^K \sum_{j=1}^K N_{ij}$ is the total number of evaluated cells ($N = 1,589$ overall; $N = 227$ per domain; smaller for intervention-type and clinical-domain strata).

Per-class F1. RoB assessment is a three-class problem ($K = 3$). Per-class metrics are computed under a one-vs-rest scheme: each study-domain cell is treated as a binary classification (“class k ” vs. “not class k ”) for each class $k \in \{\text{Low Risk, High Risk, Unclear Risk}\}$, yielding four counts:

- $\text{TP}_k = N_{kk}$: model and ground truth both assigned class k ;
- $\text{FN}_k = \sum_{i \neq k} N_{ik}$: ground truth was k , model assigned another class (positive missed);
- $\text{FP}_k = \sum_{j \neq k} N_{kj}$: model assigned k , ground truth was another class (false alarm);
- $\text{TN}_k = \sum_{i \neq k} \sum_{j \neq k} N_{ij}$: neither model nor ground truth assigned class k .

Precision (positive predictive value) and recall (sensitivity) for class k are

$$\text{Precision}_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k}, \quad \text{Recall}_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k}, \quad (2)$$

and their harmonic mean is the F1 score for class k :

$$\text{F1}_k = \frac{2 \cdot \text{Precision}_k \cdot \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}. \quad (3)$$

Per-class F1 reveals class-specific failure modes independently of overall performance.

255 **Macro-F1.** Per-class F1 is computed over the full pool of 1,589 cells; the three
 256 class-level F1 scores are then averaged with equal weight:

$$\text{Macro-F1} = \frac{1}{K} \sum_{k=1}^K \text{F1}_k, \quad K = 3. \quad (4)$$

257 Unlike Accuracy, Macro-F1 assigns equal weight to each category regardless
 258 of prevalence, penalising models that achieve high accuracy by concentrating
 259 predictions on the majority class.

260 **Cohen's κ .** A chance-corrected agreement coefficient that adjusts observed
 261 agreement p_o for the agreement expected by chance under the assumption that
 262 annotators label independently according to their observed marginal distribu-
 263 tions:

$$\kappa = \frac{p_o - p_e^\kappa}{1 - p_e^\kappa}, \quad p_e^\kappa = \sum_{k=1}^K \hat{p}_{k,\text{model}} \cdot \hat{p}_{k,\text{gt}}, \quad (5)$$

264 where $\hat{p}_{k,\text{model}} = \frac{1}{N} \sum_{j=1}^K N_{kj}$ and $\hat{p}_{k,\text{gt}} = \frac{1}{N} \sum_{i=1}^K N_{ik}$ are the marginal proportions
 265 of class k in the model predictions and ground truth, respectively (row and column
 266 sums of N divided by N). κ is reported alongside AC1 to expose their divergence
 267 under class imbalance, illustrating the prevalence paradox (see Gwet's AC1 below).

268 **Gwet's AC1.** A chance-corrected agreement coefficient robust to the preva-
 269 lence paradox [13, 14]: when marginal distributions are highly skewed, classical
 270 chance-corrected coefficients approach zero even when observed agreement is
 271 substantial. AC1 is defined as

$$\text{AC1} = \frac{p_o - p_e}{1 - p_e}, \quad p_e = \frac{1}{K-1} \sum_{k=1}^K \pi_k (1 - \pi_k), \quad (6)$$

272 where p_o is the observed proportion of agreement, $K = 3$ is the number of
 273 categories, and $\pi_k = (\hat{p}_{k,\text{model}} + \hat{p}_{k,\text{gt}})/2$ is the mean marginal proportion for
 274 category k , with $\hat{p}_{k,\text{model}}$ and $\hat{p}_{k,\text{gt}}$ the marginal proportions of class k in the model
 275 predictions and ground truth, respectively. All metrics—including AC1—are com-
 276 puted globally by pooling all 1,589 study-domain cells; domain-level values

are not aggregated. Given the pronounced class imbalance documented in Table 1—six of seven domains have Low Risk as the majority class—AC1 is more appropriate than Cohen’s κ as the primary agreement metric for this task, since κ approaches zero under heavy prevalence skew even when observed agreement is substantial (the prevalence paradox; Feinstein and Cicchetti 14). Importantly, the ground-truth annotations used here reflect published Cochrane consensus judgements rather than a second independent annotator; AC1 therefore quantifies model-to-published-standard agreement, not inter-rater reliability in the classical sense.

Error direction (LR-anchor framework). Because the three RoB classes are not ordinally exchangeable—Low Risk acts as the “safe” default that passes a study through without additional scrutiny, while High Risk and Unclear Risk both trigger closer examination—we classify non-diagonal errors relative to Low Risk. A *lenient* error occurs when the model assigns Low Risk and the ground truth is High Risk or Unclear Risk (false pass). A *cautious* error occurs when the model assigns High Risk or Unclear Risk and the ground truth is Low Risk (false flag). A *lateral* error occurs when the model confuses High Risk with Unclear Risk or vice versa. Because both non-Low-Risk categories already trigger reviewer scrutiny, lateral errors do not affect the binary pass/no-pass decision; their consequence is second-order, affecting the precision of the label (definite bias vs. insufficient information) rather than whether the study receives attention. This LR-anchor decomposition distinguishes the two first-order error directions: lenient errors risk including potentially biased studies without scrutiny (inflating evidence-base credibility), while cautious errors risk discrediting studies with genuinely low risk of bias (reducing the available evidence base). Both carry clinical costs in opposite directions; their relative severity depends on review context and the scarcity of eligible studies.

304 **2.6 Consensus Failures and Model Complementarity**

305 For each study–domain cell, we recorded whether at least one of the eight
306 models matched the ground truth. Cells where *no* model was correct are termed
307 *consensus failures* and represent cases in which all evaluated models failed
308 simultaneously, regardless of which model was consulted. The *any-correct rate*
309 (proportion of cells where at least one model succeeded) characterises the
310 degree of complementarity among models: a high any-correct rate alongside low
311 individual-model accuracy indicates that different models fail on different cells,
312 yielding non-redundant coverage across the ensemble.

313 **2.7 Stratified Analyses**

314 All primary metrics were recomputed within subgroups defined by intervention
315 type (four categories) and clinical domain (24 Cochrane Review Groups, multi-
316 label). For clinical domain, studies were included in every group to which they
317 belonged, so group sample sizes sum to more than 227.

318 **2.8 Statistical Inference**

319 All analyses are descriptive and exploratory. Confidence intervals are reported
320 to characterise estimation uncertainty and facilitate comparison across mod-
321 els and conditions; they are not interpreted as the outcome of pre-specified
322 confirmatory hypothesis tests. All 95% CIs were computed by study-clustered
323 bootstrap (5,000 iterations): the 227 studies were resampled with replacement,
324 all seven domain cells of each sampled study were retained, and the metric of
325 interest was recomputed on the resulting pseudo-sample; the 2.5th and 97.5th
326 percentiles of the bootstrap distribution define the interval. Resampling at the
327 study level preserves within-study correlation across domains (see Supporting
328 Information S7 for a comparison with Wilson score intervals). No familywise
329 error rate correction was applied: the number of comparisons across models,

domains, classes, and inference modes is large and the outcomes are correlated, making conventional multiplicity corrections either too conservative (Bonferroni) or difficult to calibrate without a pre-specified primary endpoint. Where 95% bootstrap CIs exclude zero, this is noted descriptively (“CI excludes zero”); the 95% level serves as a conventional descriptive reference threshold and carries no confirmatory implication in this exploratory context—it marks an unadjusted signal that may warrant further investigation, not a principled significance claim.

3 Results

3.1 Overall Performance

Figure 3 summarises accuracy, Macro-F1, per-class F1, and Gwet’s AC1 for all eight models across the 1,589 annotated cells, together with study-clustered bootstrap 95% confidence intervals (5,000 iterations, resampling the 227 studies—the unit of clustering—with replacement to preserve within-study correlation across the seven domains). Overall accuracy ranged from 57.6% (Gemini 2.5 Pro) to 62.5% (Claude Haiku 4.5), a spread of less than five percentage points; the 95% CI for each model spans approximately 5–6 pp, so all pairwise accuracy differences fall within sampling uncertainty and should be treated as directional. Macro-F1 confidence intervals similarly overlap across all eight models, precluding reliable rank ordering based on this metric alone. By contrast, Gemini 2.5 Flash and DeepSeek v4-Flash tied for the highest Macro-F1 (0.558) while Claude Haiku 4.5—the accuracy leader—ranked sixth (0.541). The per-class breakdown reveals that the Macro-F1 gap is driven primarily by High Risk and Unclear Risk detection: Claude Haiku 4.5 achieved the highest Low Risk F1 (0.743) but the lowest High Risk F1 (0.469) and Unclear Risk F1 (0.413) among all models. Among the three metrics, Gwet’s AC1 shows the clearest separation at the extremes: the 95% CI for Claude Haiku 4.5 ($AC1 = 0.485$; CI [0.443, 0.526]) does not overlap with those of Gemini 2.5 Pro ([0.344, 0.420]) or GPT-5.4 ([0.349, 0.426]), suggesting that Haiku’s

357 higher chance-corrected agreement is unlikely to be attributable to sampling
 358 variation alone. Consensus failures—cells where no model in the ensemble was
 359 correct—accounted for 226 of 1,589 cells (14.2%); pairwise complementarity analy-
 360 sis suggests that models make substantially non-redundant errors (Section 3.7).

Overall model performance — 95% bootstrap CIs (study-clustered, 5 000 iterations)

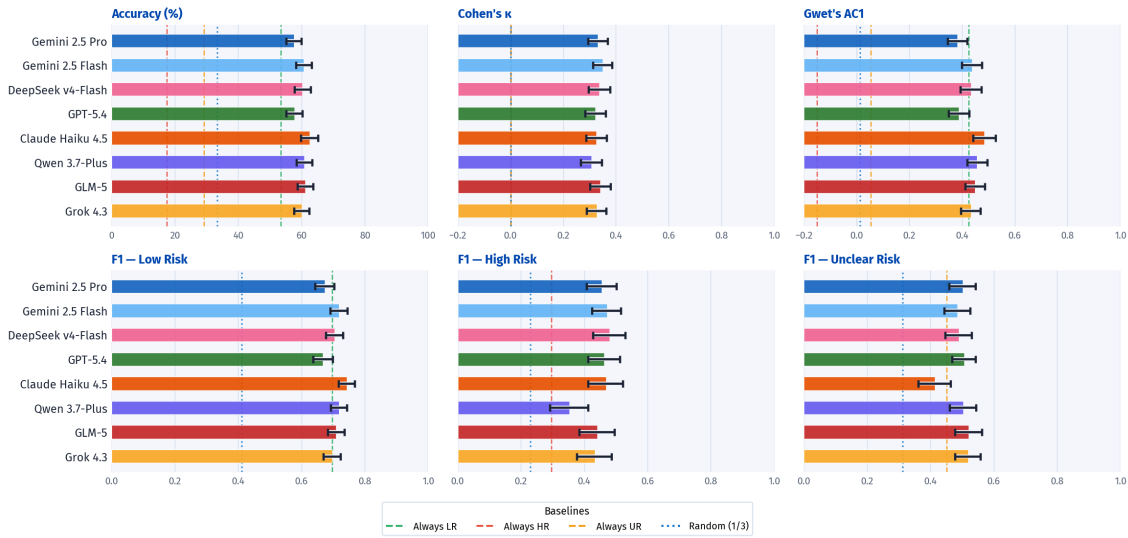


Figure 3: Overall performance on RoB 1.0 assessment for the eight models ($n = 227$ studies; 1,589 cells). Each row is one model; columns show accuracy (%), Cohen's κ , Gwet's AC1, and F1 for each risk class (F1-LR, F1-HR, F1-UR). κ and AC1 are placed side by side to expose their divergence under class imbalance (prevalence paradox). Bars represent point estimates; error bars are 95% study-clustered bootstrap confidence intervals (5,000 iterations, resampling 227 studies with replacement). Dashed vertical lines show analytical baselines: *Always LR* (green), *Always HR* (red), *Always UR* (orange), and *Random* uniform (blue dotted); baselines with F1 = 0 or $\kappa = 0$ for a given metric are not drawn. For κ and AC1, the dashed grey line marks zero (chance-level agreement). Macro-F1 is reported in Table SS1. Colours identify models consistently throughout the paper. Full numerical values are reported in Table SS1.

361 **Metric divergence.** The discrepancy between accuracy and Macro-F1 rankings
 362 reflects differential handling of the class imbalance documented in Table 1.
 363 Figure 3 makes the trade-off explicit: models with higher Low Risk F1 tend to
 364 achieve higher accuracy but lower High Risk and Unclear Risk F1, since Low Risk
 365 is the majority class in six of seven domains. Macro-F1 penalises this behaviour
 366 by weighting each class equally, shifting the ranking toward models with more
 367 balanced per-class detection. Inspection of Figure 3 also reveals that the rank

368 orderings for accuracy, AC1, and F1-LR are nearly identical across the eight models.
369 This co-linearity suggests that all three metrics are largely co-determined by Low
370 Risk performance: a model that reliably assigns Low Risk to majority-class cells
371 captures most of the variance in both accuracy and chance-corrected agreement,
372 since Low Risk prevalence (53.5%) dominates both the observed agreement
373 count and the marginal distributions that enter the AC1 correction. Macro-F1
374 disrupts this co-linearity by treating each class as equally important regardless
375 of prevalence. However, equal class weighting does not reflect the empirical
376 frequency of judgements—High Risk accounts for only 17.4% of cells (Table 1)—so
377 macro-F1 amplifies the contribution of the rarest class beyond its share of the
378 data. A prevalence-weighted F1 would instead converge toward accuracy, whereas
379 macro-F1 should be interpreted as a measure of balanced class coverage: how
380 well a model performs across *all three* nominal categories, independently of how
381 often each arises.

382 **3.2 Performance by RoB Domain**

383 Domain-level accuracy (Figure 4; Table SS2) revealed two tiers of difficulty. **D1** (se-
384 quence generation) and **D2** (allocation concealment) were the most accurately
385 assessed domains, with mean accuracy of 75.3% and 72.7%, respectively; all mod-
386 els exceeded 70% accuracy on D1. **D3** (blinding of participants) showed high
387 variance across models (49.8%–72.2%), with Gemini 2.5 Flash and Claude Haiku 4.5
388 outperforming others by a margin of more than 5 percentage points.

389 **D7** (other sources of bias) was the most challenging domain for all models,
390 with accuracy ranging from 24.2% (Gemini 2.5 Pro) to 55.5% (Qwen 3.7-Plus)—
391 a spread of 31 percentage points within a single domain. Notably, models at
392 opposite ends of the D7 accuracy range adopted diametrically opposed prediction
393 strategies: Gemini 2.5 Pro assigned High Risk to over 93% of D7 cells (recall 0.94
394 for High Risk; recall 0.18 for Low Risk), whereas Qwen 3.7-Plus assigned Low
395 Risk to 88% of D7 cells (recall 0.88 for Low Risk; recall 0.06 for High Risk). Both

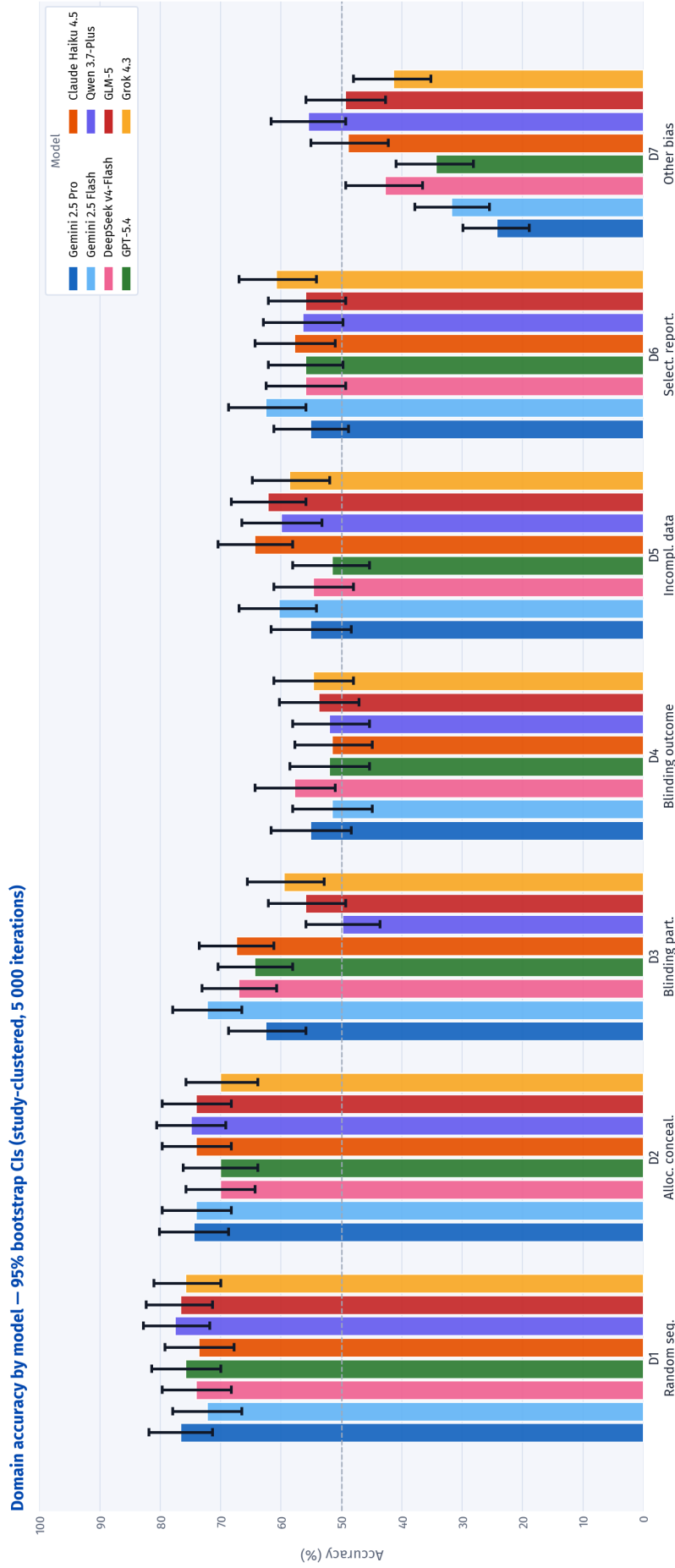


Figure 4: Per-domain accuracy by model (95% bootstrap CIs; study-clustered, 5,000 iterations, resampling 227 studies with replacement). Each group of eight bars corresponds to one RoB domain; colours identify models as shown in the legend. The dashed horizontal line marks 50% (majority-class baseline). All analyses are exploratory. See Figure 5 for the breakdown by Macro-F1, per-class F1, and Gwet's AC1.

396 strategies yielded similarly poor Macro-F1 in D7 (≤ 0.37 for both), demonstrating
397 that high accuracy on a single class does not translate to overall competence.

398 Consensus failures were not uniformly distributed across domains: the con-
399 sensus failure rate was lowest for D2 (8.8% of its 227 cells) and highest for D4
400 (22.0%), consistent with the greater class balance and annotation difficulty in D4
401 relative to D2.

402 The domain-level pattern identifies two structurally distinct types of difficulty.
403 **D4** (blinding of outcome assessors) exhibits systematic, directionally consistent
404 errors: across all eight models, the large majority of errors are lenient—models
405 assign Low Risk when the ground truth is High Risk (see Figure 6)—a pattern
406 present across all intervention types. In drug trials in particular, models appear
407 to conflate the objectivity of the outcome measure with the adequacy of blinding
408 of the outcome assessor, defaulting to Low Risk when no explicit blinding failure
409 is stated. **D7** (other sources of bias), by contrast, is characterised by inter-model
410 divergence: models adopt idiosyncratic strategies—some biased toward High Risk,
411 others toward Low Risk—so that errors are distributed roughly equally between
412 lenient and cautious directions. The wide accuracy spread across models in D7
413 (31 pp) and its near-zero chance-corrected agreement thus reflect not a shared
414 systematic error but genuine criterion ambiguity; the residual difficulty in D7 is
415 unlikely to be resolved by prompt refinement alone.

416 **Per-class breakdown.** Figure 5 disaggregates performance into the three
417 constituent classes across all domains. Across all models, the ordering $F1\text{-}LR \gg F1\text{-}UR \geq F1\text{-}HR$
418 holds as a near-universal pattern, closely mirroring the
419 prevalence ordering (53.5%, 29.1%, and 17.4%, respectively; Table 1). The sole
420 exception is Claude Haiku 4.5, which inverts the UR–HR ordering ($F1\text{-}HR = 0.469 >$
421 $F1\text{-}UR = 0.413$): Haiku is more willing to assign High Risk when text signals a
422 methodological concern, at the cost of under-using Unclear Risk as an epistemic
423 hedge—consistent with its highest $F1\text{-}LR$ (0.743) and lowest $F1\text{-}UR$ (0.413) in the
424 benchmark.

425 **Low Risk** detection was strong and consistent across models in D1 and D2
426 ($F1 > 0.75$ for all models), but deteriorated in D7: most models reported $F1$ -LR
427 below 0.60 there, and Gemini 2.5 Pro collapsed to $F1 = 0.29$ as a consequence of
428 its High Risk-biased prediction strategy.

429 **High Risk** detection was reliable only in D3 ($F1$: 0.47–0.79), the single domain
430 where High Risk cases outnumbered Low Risk in the ground truth (103 vs. 74;
431 Table 1). The wide bootstrap CIs for $F1$ -HR in D1 and D2 (Figure 5) reflect the small
432 number of High Risk cases in those domains (D1: $n = 5$; D2: $n = 10$), such that a
433 single misclassified study shifts the $F1$ estimate substantially under resampling.
434 In D6 (selective outcome reporting), four models (Gemini 2.5 Flash, GPT-5.4,
435 Claude Haiku 4.5, and Qwen 3.7-Plus) reported High Risk $F1 = 0.00$, corresponding
436 to zero true-positive High Risk predictions across all 227 studies: High Risk is the
437 least prevalent class in D6 (25 of 227 cases, 11%), and these models systematically
438 defaulted to Low Risk or Unclear Risk in the absence of explicit evidence of
439 selective reporting.

440 **Unclear Risk** was the most consistently under-detected class. Mean recall
441 fell below 0.20 in D3, D5, and D7 across all models. The worst single result
442 was Claude Haiku 4.5 in D5: recall = 0.04 (2 of 45 Unclear Risk cases correctly
443 identified). The most accurately detected Unclear Risk domain was D2, where
444 all models achieved $F1 > 0.70$, consistent with the near-equal prevalence of Low
445 Risk and Unclear Risk in that domain (113 vs. 104). The pattern of under-detection
446 reflects an asymmetric decision behaviour: in D3, D4, D5, and D7—where the
447 ground-truth Unclear Risk rate ranges from 20% to 29%—all models committed
448 to definitive Low Risk or High Risk judgements more often than the Cochrane
449 reviewer, even in cases judged as informationally insufficient. In D2 and D6 the
450 pattern reverses, with models over-using Unclear Risk relative to the ground truth.
451 This bidirectional discrepancy indicates that models calibrate their uncertainty
452 threshold differently across domains; reducing false positives for Unclear Risk
453 in the blinding domains (D3–D4) is a necessary condition for improving High

454 Risk recall there. One prompt-design factor may contribute to the over-use of
455 Unclear Risk in D6: the prompt specifies a *compound* criterion for that domain
456 (“no protocol/registry **and** insufficient information”), requiring both conditions
457 simultaneously, whereas all other domains require only “*insufficient information*”
458 to trigger Unclear Risk. This asymmetry may lower the Unclear Risk threshold in
459 D6 relative to other domains, resulting in higher Unclear Risk assignment rates
460 and correspondingly lower Low Risk rates.

461 **3.3 Error Direction: Leniency and Caution**

462 Among cells where models erred, errors were not directionally neutral. Figure 6
463 decomposes errors into three categories under the LR-anchor framework: *lenient*
464 (Low Risk assigned when High Risk or Unclear Risk was correct), *cautious* (High
465 Risk or Unclear Risk assigned when Low Risk was correct), and *lateral* (High Risk
466 and Unclear Risk confused with each other—the study remains flagged but with
467 a mislabelled severity).

468 **D4** (blinding of outcome assessors) exhibited the most pronounced and con-
469 sistent leniency (50–74% across intervention types), with drug/medication trials
470 showing the highest lenient fraction (74%) and lateral confusion remaining low
471 ($\leq 35\%$). Models appear to conflate the objectivity of the outcome measure with
472 adequate assessor blinding, defaulting to Low Risk when no explicit blinding
473 failure is stated.

474 **D3** (blinding of participants and personnel) showed both high leniency and
475 the highest lateral fraction in the benchmark. While leniency was elevated
476 across all intervention types (48–76%), the lateral fraction was highest for non-
477 pharmacological interventions (42%) and medical devices (24%), indicating that
478 models often recognised a blinding concern without resolving whether it war-
479 ranted High Risk or Unclear Risk—a distinction that turns on reporting quality
480 rather than intervention feasibility.

481 **D1** (sequence generation) and **D2** (allocation concealment) showed a pre-

482 dominantly *cautious* error profile: when models erred on these domains, they
483 over-flagged Low Risk studies as High Risk or Unclear Risk (55–87% cautious
484 in D1; 49–100% in D2). This reversal relative to D3–D4 is consistent with the
485 explicit, formulaic language used to describe randomisation procedures—models
486 rarely miss a clear statement of adequate randomisation and instead raise false
487 concerns when reports are terse.

488 **D6** (selective outcome reporting) was also predominantly cautious (48–72%),
489 reflecting a tendency to flag concerns in the absence of a registered protocol
490 even when the ground-truth judgement was Low Risk.

491 **D7** (other sources of bias) showed no dominant direction, with roughly bal-
492 anced lenient and cautious fractions and a lateral component of 12–21%—confirming
493 the idiosyncratic, criterion-ambiguous character of errors in this domain noted
494 in Section 3.2.



Figure 5: Per-domain performance by model across five metrics: Cohen's κ , F1 for Low Risk, Unclear Risk, and High Risk, and Gwet's AC1 (95% bootstrap CIs; study-clustered, 5,000 iterations, resampling 227 studies with replacement). Each panel shares the same domain grouping on the x -axis; bars are point estimates; error bars are 95% CIs. Short horizontal segments show domain-specific analytical baselines (*Always LR* green, *Always HR* red, *Always UR* orange), computed from each domain's class prevalence; the blue dotted line is the *Random* uniform baseline (constant 1/3 across domains). Baselines with F1 = 0 for a given metric are not drawn. For κ and AC1, the dashed grey line marks zero (chance-level agreement). Macro-F1 by domain is reported in Table SS3. Colours identify models consistently throughout the paper.

Error direction by domain • intervention type (LR-anchor framework)

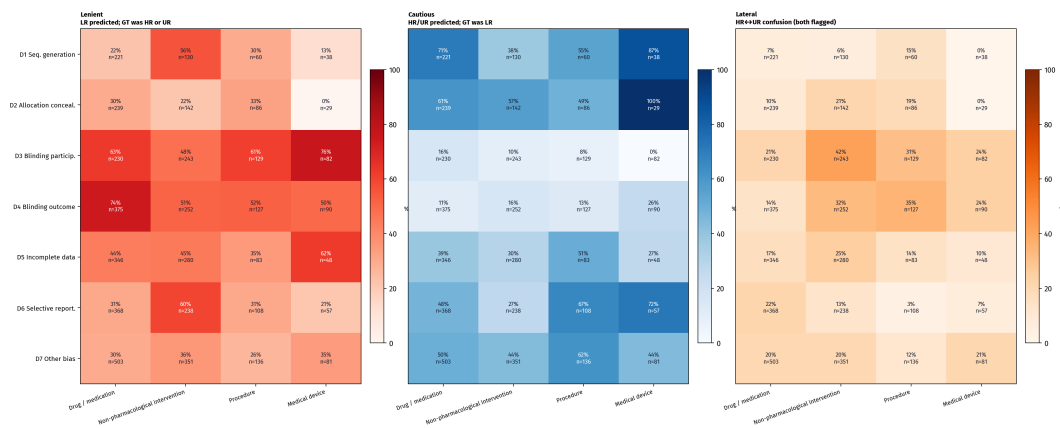


Figure 6: Error direction by RoB domain and intervention type under the LR-anchor framework (three panels, pooled across all eight models). **Lenient** (red): model assigns Low Risk when ground truth was High Risk or Unclear Risk—the study passes unscrutinised. **Cautious** (blue): model assigns High Risk or Unclear Risk when ground truth was Low Risk—a valid study is flagged for unnecessary review. **Lateral** (orange): model confuses High Risk with Unclear Risk or vice versa—the study remains in the flagged pile but with a mislabelled severity. The three percentages sum to 100% in each cell; n = total errors for that domain–intervention combination (identical across all three panels). Grey cells indicate no errors in that combination.

495 3.4 Stratification by Intervention Type

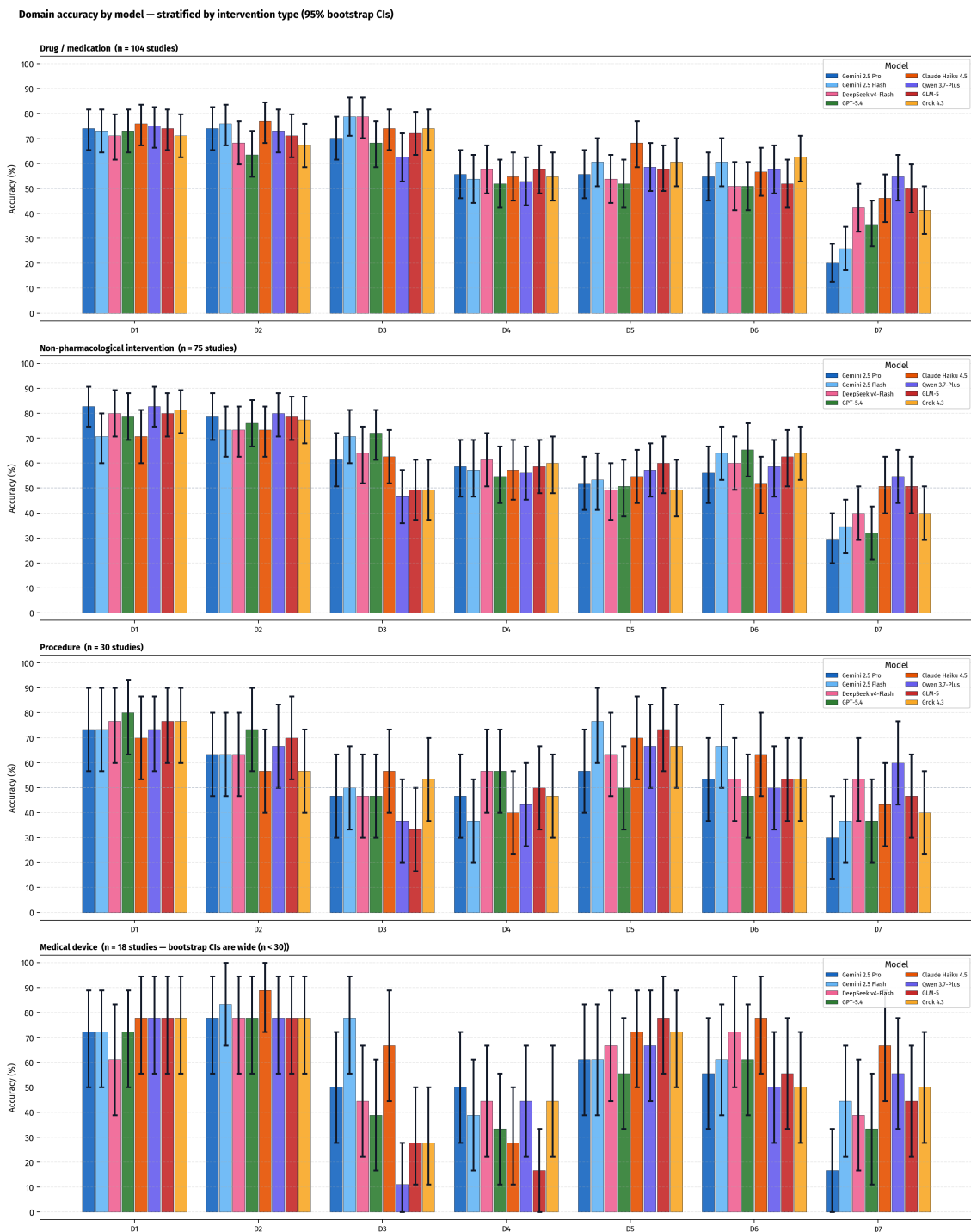


Figure 7: Per-domain accuracy by model, stratified by intervention type (95% bootstrap CIs; study-clustered, 5,000 iterations). Panels from top to bottom: drug/medication ($n = 104$), non-pharmacological intervention ($n = 75$), procedure ($n = 30$), and medical device ($n = 18$). Bootstrap CIs for the two smaller strata are wide and should be interpreted with caution; see Supporting Information S4 for numeric estimates.

Mean Macro-F1 across all models ranked intervention types as follows: non-pharmacological intervention (0.572, $n = 75$ studies), drug/medication (0.538, $n = 104$), medical device (0.531, $n = 18$), and procedure (0.510, $n = 30$). The 6.2-point gap between non-pharmacological interventions and procedures was consistent in direction across most models; no single model reversed this ordering. However, bootstrap CIs for the two smaller strata (medical device, $n = 18$; procedure, $n = 30$) are wide (see Figure 7), so this pattern should be interpreted as a descriptive observation rather than a reliable rank ordering pending replication in larger strata. Individual model rankings within each intervention type are shown in Figure 7 and provided in full in Table SS4.

The interaction between intervention type and blinding assessment reveals a specific failure mode. In **D3** (blinding of participants and personnel), accuracy for drug/medication trials substantially exceeded that for device and procedural trials; in the latter, the ground-truth High Risk prevalence is highest—because physical concealment of the intervention is frequently impossible—yet models assigned Low Risk to the majority of cells, producing leniency rates above 65% (Figure 6). In **D4** (blinding of outcome assessors), the leniency bias reached its maximum in drug trials (exceeding 80%), where models appear to assume that objective outcome measurement implies adequate assessor blinding. These findings converge on a specific prompt deficiency: the current zero-shot prompt does not distinguish between trial types when applying blinding criteria. Models apply a “blinding is feasible and likely” default regardless of the physical feasibility of concealment, an error that a single conditional rule in the prompt (e.g., flagging as High Risk by default when concealment is structurally impossible and no blinding mechanism is described) would substantially reduce.

3.5 Stratification by Clinical Domain

Among clinical domains with at least five studies, mean Macro-F1 across models ranged from 0.425 (Infectious Disease, $n = 8$) to 0.680 (Orthopaedics & Trauma,

$n = 7$)—a 25.5-point spread (Figure 8). The four highest-performing domains were Orthopaedics & Trauma (0.680), Endocrine & Metabolic (0.638, $n = 10$), Rheumatology (0.637, $n = 5$), and Wounds (0.628, $n = 5$). The four lowest-performing domains were Infectious Disease (0.425), Reproductive & Sexual Health (0.437, $n = 8$), Genetic Disorders (0.473, $n = 9$), and Mental Health (0.481, $n = 11$). Complete results for all 24 Cochrane Review Groups are provided in Table SS5.

These results are hypothesis-generating rather than confirmatory. Group sizes of 5–11 studies at the extremes preclude reliable rank-ordering across domains, and multi-label tagging makes groups non-exclusive (some reviews are cross-listed under multiple Cochrane Review Groups). The pattern is nonetheless qualitatively consistent with an interpretable explanation: domains characterised by standardised reporting (orthopaedic interventions and metabolic trials typically follow structured protocols with explicit randomisation and blinding procedures) show higher Macro-F1, while domains with heterogeneous or complex study designs (Infectious Disease, Reproductive & Sexual Health) present harder RoB characterisation challenges for models trained on general text. These observations motivate targeted future evaluation on larger domain-specific corpora.

3.6 Ground-Truth Uncertainty and Inter-Annotator Agreement

The benchmark treats published Cochrane annotations as the reference standard, but those annotations carry their own measurement error. Even among formally trained reviewers applying the same protocol, Hartling et al. [1] reported $\kappa = 0.79$ for D1 but only $\kappa = 0.24$ – 0.37 for the remaining six domains, with agreement between independent consensus pairs falling further to $\kappa = 0.05$ – 0.09 in several domains. Agreement also varies substantially between distinct Cochrane review teams evaluating the same trials [4]. These findings contextualise model performance against a noisy reference standard rather than a noise-free human ceiling, and motivate domain-level rather than aggregate interpretation of re-

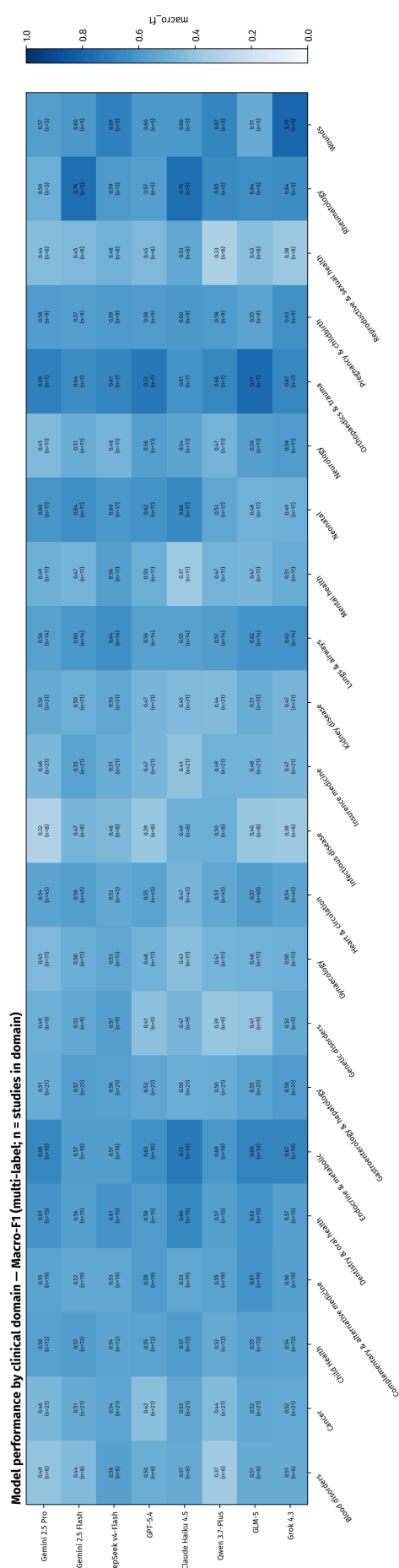


Figure 8: Macro-F1 by clinical domain and model (domains with ≥ 5 studies; n = number of studies in each domain, multi-label). Colour scale: 0–1.

sults. Figure 9 places model performance in this context by displaying the full 9×9 label-agreement matrix across Ground Truth and all eight models.

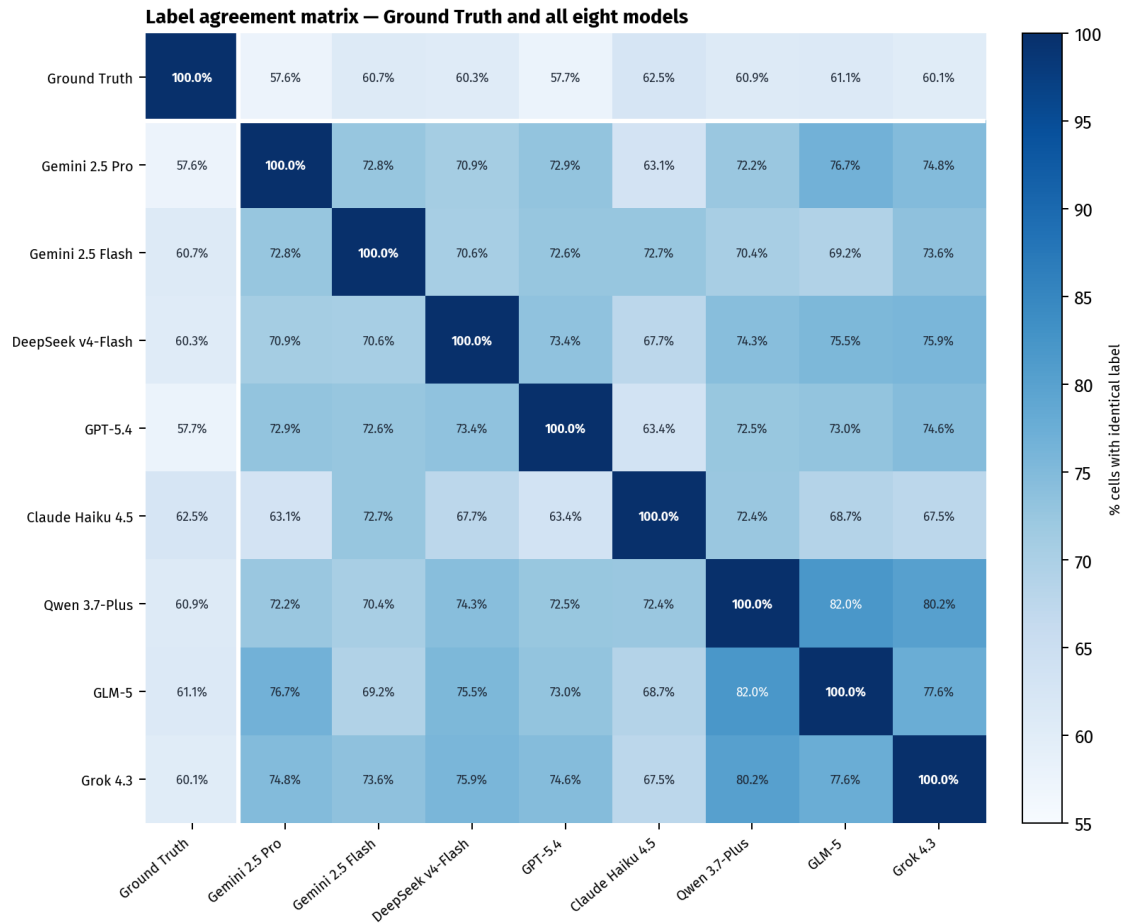


Figure 9: Label agreement matrix for Ground Truth and all eight models ($n = 1,589$ cells). Cell (i, j) = percentage of cells in which annotator i and annotator j assigned the *same* label, irrespective of correctness. The diagonal is 100% by definition. The white lines isolate the Ground Truth row and column. Colour scale: 55–100% (Blues).

The Ground Truth row (first row) reproduces the known model accuracies (57.6%–62.5%), representing agreement between each model and the Cochrane reference. All inter-model cells (the lower-right 8×8 block) exceed these values, ranging from 63.1% (Claude Haiku 4.5 and Gemini 2.5 Pro) to 82.0% (Qwen 3.7-Plus and GLM-5). These pairwise statistics are descriptive: no bootstrap confidence intervals were computed for them. Because cells from the same study share textual context and are correlated across domains, the effective sampling unit is the study ($n = 227$), not the cell ($n = 1,589$); each pairwise proportion therefore

562 carries an approximate 95% uncertainty of ± 6 pp ($SE \approx \sqrt{p(1-p)/227}$, evaluated
563 at $p \approx 0.70$). Differences smaller than this margin should not be over-interpreted;
564 the patterns noted below reflect the overall structure of the matrix rather than
565 fine-grained rankings between adjacent cells. This systematic gap—models agree
566 with each other more than they agree with the ground truth—indicates that
567 the eight models share systematic prediction tendencies that diverge from the
568 Cochrane standard in coordinated ways. It does not imply that the models are
569 collectively more reliable than the ground truth; rather, it reflects shared training
570 priors that cause convergent errors.

571 The magnitude of the gap is interpretable in light of the known reference
572 noise: if Cochrane reviewer pairs themselves agree at ≈ 60 – 79% (the $\kappa = 0.24$ – 0.37
573 range, converted to expected agreement under the observed class distribution),
574 then the inter-model agreement range of 63 – 82% is broadly comparable to or
575 modestly above human inter-rater reliability on this task. The ground-truth
576 column therefore represents one reference annotator among many possible
577 ones, and the model accuracy figures (57 – 63%) should be interpreted as lower
578 bounds on the models’ latent capability rather than definitive measurements of
579 their performance against a noise-free standard.

580 **3.7 Consensus Failures and Model Complementarity**

581 Table 3 summarises the full distribution of per-cell model agreement across the
582 1,589 study-domain cells. The distribution is markedly bimodal: the two largest
583 individual categories are *all models correct* (499 cells, 31.4%) and *consensus*
584 *failure* (226 cells, 14.2%), with the intermediate categories together accounting
585 for the remaining 54.4%. This bimodality suggests that cell difficulty is largely
586 an intrinsic property of the item rather than the model: most cells are either
587 solvable by all frontier models evaluated here, or intractable for all of them
588 given the prompt context.

Table 3: Distribution of per-cell model agreement across the 1,589 study-domain cells (8 models). *Majority* and *minority* thresholds are defined relative to $n = 8$; see Section 2.6 for the generalised formula.

Category	N	%
All correct (8/8)	499	31.4
Majority (5–7/8)	418	26.3
Split (4/8)	96	6.0
Minority (1–3/8)	350	22.0
None correct (0/8)	226	14.2
Total	1,589	100.0

Table 4 disaggregates the distribution by domain. D1 and D2 are the easiest: 54.2% and 46.3% of their cells achieve full eight-model consensus, and consensus failure rates are 11.0% and 8.8%, respectively. D7 (other sources of bias) stands apart in a different way: only 4.0% of its cells are answered correctly by all models, and 45.8% are answered correctly only by a minority (1–3 models), reflecting a domain where models systematically diverge in their judgements. D4 and D5 present a third pattern: their consensus failure rates (22.0% and 19.8%) are the highest in the corpus—higher even than D7 (12.3%)—indicating that when models err in D4/D5 they tend to do so collectively, whereas in D7 errors are more dispersed across models.

Table 4: Per-cell agreement distribution by domain (row percentages; $n = 227$ per domain).

Domain	All (8/8)	Majority (5-7)	Split (4)	Minority (1-3)	None (0)
	%	%	%	%	%
D1	54.2	22.5	1.3	11.0	11.0
D2	46.3	27.3	2.6	15.0	8.8
D3	33.5	28.2	7.0	17.6	13.7
D4	32.2	16.3	6.6	22.9	22.0
D5	30.4	25.1	7.9	16.7	19.8
D6	19.4	37.4	6.2	25.1	11.9
D7	4.0	27.3	10.6	45.8	12.3

599 Stratifying by ground-truth class (Table 5) reveals a clear asymmetry. *Low risk*
600 cells are classified correctly by all eight models in 44.4% of cases, and only 4.9%
601 are consensus failures. *Unclear risk* cells are the hardest: 27.0% are consensus
602 failures—no model assigns the correct label—exceeding even *high risk* (21.4%).
603 This result is consistent with the inherent difficulty of the *unclear risk* category,
604 which requires inferring the *absence* of adequate reporting rather than identifying
605 a positive signal of bias.

Table 5: Per-cell agreement distribution by ground-truth class (row percentages).

Class	n	All (8/8)	Majority (5-7)	Split (4)	Minority (1-3)	None (0)
		%	%	%	%	%
Low risk	850	44.4	25.9	5.1	19.8	4.9
High risk	276	13.8	27.5	9.8	27.5	21.4
Unclear risk	463	18.1	26.3	5.6	22.9	27.0

606 Figure 10 shows the pairwise intersection matrix: cell (i, j) gives $P(A_i \cap A_j)$,
607 the percentage of the 1,589 cells in which *both* model i and model j produced the

correct judgement. The diagonal entries report each model’s overall accuracy
 $(P(A_i))$.

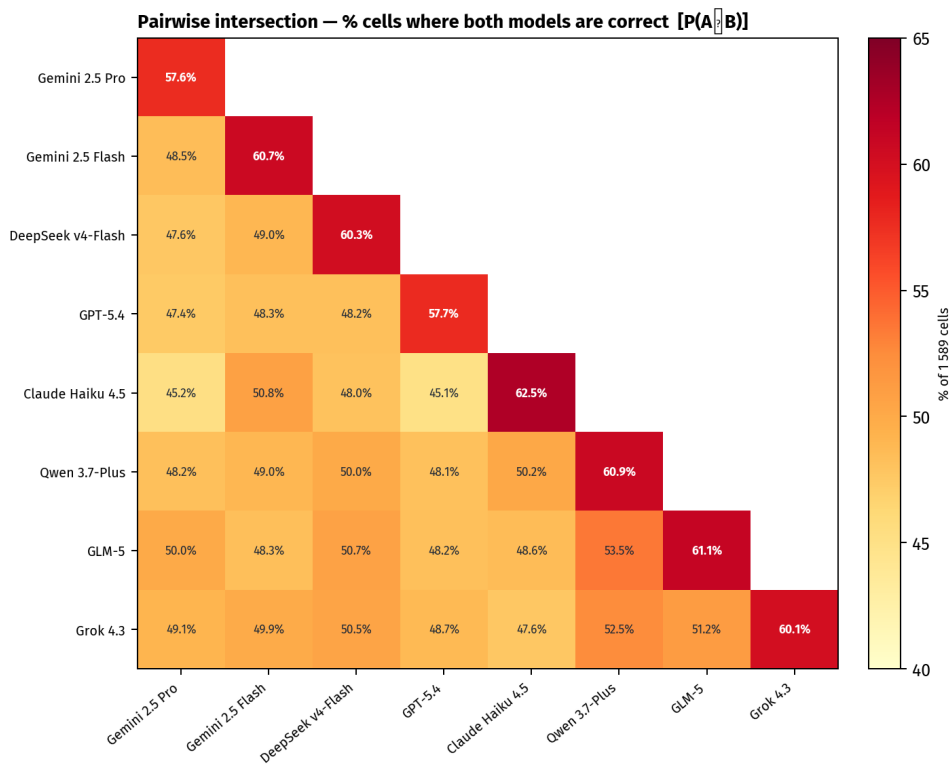


Figure 10: Pairwise intersection matrix. Cell $(i, j) = P(A_i \cap A_j)$: percentage of the 1,589 study-domain cells in which both models were correct. Diagonal entries (bold) = individual model accuracy. Colour scale: 40–65% (YlOrRd); deeper red = greater overlap in correct cells. Upper triangle omitted by symmetry.

Off-diagonal intersection values range from 45.1% (Claude Haiku 4.5 and GPT-5.4) to 53.5% (Qwen 3.7-Plus and GLM-5), all substantially below the respective diagonal entries (57.6%–62.5%). The gap between diagonal and off-diagonal reflects that no two models share all of their correct cells: a non-trivial fraction of each model’s successes are unique to that model. Expressing overlap as the Jaccard similarity $J(A, B) = P(A \cap B) / P(A \cup B)$ —the fraction of cells correct for *at least one* model that are correct for *both*—yields values between 60.0% and 78.1% across the 28 pairs. The highest Jaccard, Qwen 3.7-Plus and GLM-5 ($J = 78.1\%$, $P(A \cap B) = 53.5\%$), indicates that nearly four in five cells that either model answers correctly are answered correctly by both, confirming this as the most redundant pair. The lowest Jaccard, Claude Haiku 4.5 and GPT-5.4

621 ($J = 60.0\%$, $P(A \cap B) = 45.1\%$), indicates that two in five jointly-correct cells
622 belong exclusively to one model—the highest non-redundancy in the ensemble
623 and a direct predictor of the high complementarity entry visible in Figure 11.

624 Figure 11 shows the full pairwise complementarity matrix: cell (i, j) gives the
625 percentage of the 1,589 cells in which model i erred and model j was correct.
626 Values range from 7.4% to 17.4%, with all off-diagonal entries exceeding 7%,
627 confirming that no two models in the ensemble are redundant.

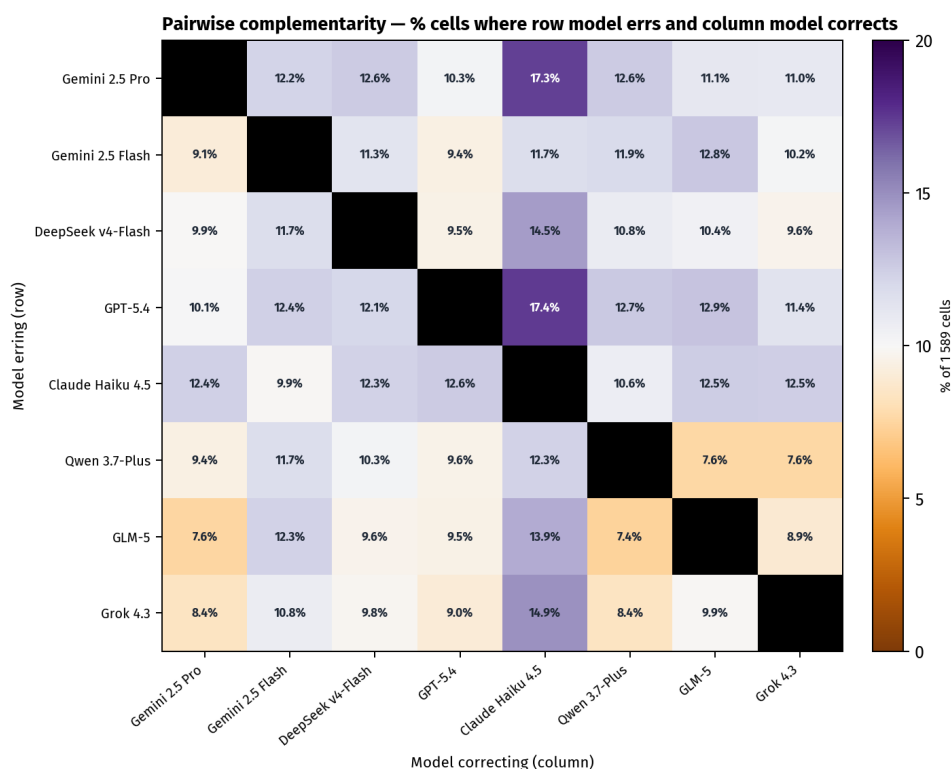


Figure 11: Pairwise complementarity matrix. Cell (i, j) = percentage of the 1,589 study-domain cells in which the row model erred and the column model was correct. Colour scale: 0–20% (PuOr); purple = high complementarity, orange = low complementarity; grey diagonal = undefined. Full numeric values are reported in Table SS6.

628 The most salient feature of Figure 11 is the Claude Haiku 4.5 column: it is the
629 darkest column overall, meaning that when any other model errs, Haiku recovers
630 those cells at the highest rate (12.3%–17.4%). This is the direct complement of
631 its accuracy profile—Haiku leads on overall accuracy and AC1 by concentrating
632 predictions on the majority class, which makes its correct cells less likely to be
633 errors for other models, but also makes its errors structurally distinct from theirs.

634 The two highest individual entries in the matrix—GPT-5.4 → Haiku (17.4%) and
 635 Gemini 2.5 Pro → Haiku (17.3%)—correspond to the two models with the lowest
 636 overall accuracy; their larger error sets overlap least with Haiku’s. The same
 637 structural distinctiveness is visible in the agreement matrix (Figure 9): Haiku
 638 has the lowest mean inter-model agreement across the ensemble (67.9%, versus
 639 71.7%–74.9% for all other models), driven by consistently low agreement with
 640 Gemini 2.5 Pro and GPT-5.4 (63–64%)—the same dyads that top the complemen-
 641 tarity column. High complementarity and low overall agreement are two sides
 642 of the same coin: because Haiku assigns labels differently from the ensemble
 643 cluster, it agrees with other models less often irrespective of correctness, but
 644 those disagreements are disproportionately cases where Haiku is right and the
 645 other model is wrong. At the opposite extreme, the Qwen 3.7-Plus ↔ GLM-5 pair
 646 shows the lowest bidirectional complementarity (7.4% and 7.6%), indicating the
 647 most overlapping error profiles of any pair in the ensemble. Grok 4.3 also shows
 648 low complementarity as a *column* model for the Qwen and GLM-5 rows (7.6% and
 649 8.9%), suggesting that these three models share a similar error structure that is
 650 not well covered by the others. This pattern is reinforced by the agreement matrix
 651 (Figure 9): the three highest pairwise inter-model agreements—Qwen 3.7-Plus /
 652 GLM-5 (82.0%), GLM-5 / Grok 4.3 (80.2%), and Qwen 3.7-Plus / Grok 4.3 (77.6%)—are
 653 all concentrated within this trio; the trio’s mean agreement (79.9%) exceeds the
 654 mean of cross-trio pairs by 7.2 pp. Taken together, the matrix indicates that
 655 Claude Haiku 4.5, Gemini 2.5 Pro, and GPT-5.4 form the most complementary triad:
 656 their pairwise entries are uniformly above 10%, and combining any two of them
 657 recovers more than a fifth of all cells that either individually misses.

658 **3.8 Inference Mode Comparison**

659 Table 6 compares per-domain and single-call inference modes for the two models
 660 evaluated under both conditions.

661 The overall picture from this two-model pilot comparison (Table 6) is one of

Table 6: Per-domain vs. single-call inference mode comparison for Gemini 2.5 Flash and DeepSeek v4-Flash ($n = 227$ studies; 1,589 cells each). F1 scores are computed globally across all cells (identical methodology to Table SS1); per-domain values reproduce the corresponding entries in Table SS1. Δ : single-call minus per-domain; 95% paired bootstrap CI (study-clustered, 5,000 iterations) shown in brackets. †: CI excludes zero (unadjusted for multiplicity; all analyses exploratory).

Model	Accuracy (%)	Macro-F1	F1-LR	F1-HR	F1-UR	Gwet's AC1
<i>Gemini 2.5 Flash</i>						
Per-domain	60.7	0.558	0.718	0.474	0.484	0.437
Single-call	59.8	0.548	0.715	0.490	0.440	0.425
Δ	-0.9	-0.010	-0.003	+0.016	-0.044†	-0.012
	[-2.7, +1.0]	[-0.030, +0.011]	[-0.019, +0.014]	[-0.013, +0.051]	[-0.079, -0.009]	[-0.038, +0.015]
<i>DeepSeek v4-Flash</i>						
Per-domain	60.3	0.558	0.704	0.480	0.489	0.433
Single-call	62.6	0.576	0.730	0.475	0.523	0.466
Δ	+2.3	+0.018	+0.026†	-0.005	+0.034	+0.033
	[-0.1, +4.7]	[-0.007, +0.044]	[+0.004, +0.047]	[-0.049, +0.041]	[-0.003, +0.072]	[-0.001, +0.066]

small differences with wide uncertainty: of 12 Δ values across both models and six metrics, ten have 95% CIs that include zero. The two whose CIs exclude zero are a drop in Unclear Risk F1 for Gemini 2.5 Flash (-0.044 ; CI $[-0.079, -0.009]$, †) and a gain in Low Risk F1 for DeepSeek v4-Flash ($+0.026$; $[+0.004, +0.047]$, †); both are class-specific and do not translate into differences in accuracy, Macro-F1, or AC1, all of which remain within their respective uncertainty intervals. The accuracy differences (-0.9 pp for Flash, $+2.3$ pp for DeepSeek), while pointing in opposite directions, are each compatible with zero change given the bootstrap CIs ($[-2.7, +1.0]$ and $[-0.1, +4.7]$ respectively).

Domain-level results reinforce this picture. D7 (Other Bias), the domain for which joint assessment was hypothesised to confer the greatest benefit through shared contextual reasoning, showed negligible differences under both modes (± 1 pp)— the most direct evidence that collapsing seven domain assessments into a single call does not meaningfully improve the handling of the hardest domain. The opposing D3 pattern (Flash: -0.4 pp; DeepSeek: -3.6 pp) and the concentration of DeepSeek gains in D5 ($+10.2$ pp) and D2 ($+4.4$ pp) suggest that whatever domain-level shifts occur are idiosyncratic to each model rather than reflecting a systematic benefit of joint assessment.

Single-call outputs were systematically shorter (DeepSeek: 400→214 chars;

Gemini 2.5 Flash: 530→354 chars), with a corresponding decrease in hedging expressions (e.g., “unclear”, “not reported”, “cannot determine”) in free-text justifications, consistent with a more compressed generation regime under the single-call prompt. The practical implications of this output compression appear limited and model-dependent: it did not translate into a consistent performance benefit or detriment across either model at the aggregate level.

4 Conclusions

Eight state-of-the-art LLMs assessed RoB 1.0 for 227 randomised controlled trials in a zero-shot, single-agent setting, addressing whether the reliability shortfall documented in prior work is uniform across the RoB 1.0 structure or varies predictably by domain and intervention type. Overall accuracy ranged narrowly from 57.6% to 62.5%, but Macro-F1 and Gwet’s AC1 revealed meaningful differences in how models handled the three nominal classes. Models with the highest accuracy achieved it partly by concentrating predictions on majority-class labels, a strategy that inflates accuracy whilst suppressing detection of High Risk and Unclear Risk judgements—the very categories most consequential for evidence appraisal. Macro-F1 and Gwet’s AC1 are therefore more informative primary metrics than accuracy for this task.

Unclear Risk was the most consistently under-detected class across all models and domains, with recall below 0.20 in three of seven domains. This systematic failure reflects the epistemic character of the category—it requires the model to recognise that key methodological information is absent, a more complex inference than identifying a positive or negative methodological signal in the text.

Domain-level and stratification analyses revealed structured heterogeneity in performance. Sequence generation (D1) and allocation concealment (D2) were reliably assessed across all models, whereas other sources of bias (D7)

708 showed the greatest inter-model accuracy spread and blinding of outcome as-
709 sessors (D4) had the highest consensus failure rate (22.0%—no model correct
710 in 50 of 227 cells). Clinical domains with clear, domain-specific methodological
711 criteria (Orthopaedics & Trauma, Endocrine & Metabolic) supported higher Macro-
712 F1 than domains with more heterogeneous study designs (Infectious Disease,
713 Reproductive & Sexual Health).

714 Consensus failures—cells where no model was correct—accounted for 14.2%
715 (226 of 1,589 cells) and were concentrated in D4 and D5, the two most ambigu-
716 ous domains. These cases identify the boundaries of current frontier model
717 capability under this evaluation design and may reflect genuine annotation un-
718 certainty as much as model failure. Pairwise complementarity analysis showed
719 that 7.4%–17.4% of cells are correctable by substituting any one model for another,
720 confirming that the eight models make substantially non-redundant errors.

721 The comparison between per-domain and single-call inference modes (Sec-
722 tion 3.8) produced small differences that do not collectively support a perfor-
723 mance advantage for either strategy. Ten of twelve Δ values across both models
724 and six metrics had 95% CIs spanning zero; the two exceptions were class-specific
725 effects of limited practical magnitude (Unclear Risk F1 for Flash: -0.044 ; Low
726 Risk F1 for DeepSeek: $+0.026$) that did not propagate to aggregate metrics. Most
727 notably, D7 (Other Bias)—the domain for which joint contextual assessment
728 was hypothesised to be most beneficial—showed negligible differences under
729 both modes (± 1 pp), indicating that its difficulty is intrinsic to the vagueness of
730 the criterion rather than a consequence of fragmented inference calls. Taken
731 together, the preliminary evidence from this two-model comparison does not
732 identify a meaningful performance rationale for preferring single-call over per-
733 domain assessment; the primary practical distinction between the two modes
734 is computational (single-call reduces inference calls by $7\times$) rather than one of
735 accuracy.

736 Reframing High Risk detection as a binary classification task (High Risk vs. not

High Risk) highlights the clinical stakes of the leniency bias documented above. Under this framing, the best-performing models for High Risk recall detected fewer than two thirds of High Risk cells (F1-HR: 0.455–0.480 for the top-ranked models), meaning roughly one in three cells that should trigger reviewer scrutiny passed undetected. Both error directions carry clinical costs: a false negative (lenient—biased study classified as acceptable and included unquestioned in a meta-analysis) risks inflating pooled effect estimates, while a false positive (cautious—a low-risk study flagged as problematic) risks unjustifiably discrediting or excluding credible evidence, a concern that grows when the pool of eligible studies is small. The relative severity depends on review context; however, tools intended to assist real-world screening workflows would need to make this trade-off explicit rather than leaving it to the zero-shot prompt’s implicit defaults.

The cell-level concordance partition offers a practical framework for directing qualitative review effort. Cells in which all models converge on the correct answer require no additional attention. Cells in which all models converge on the *same wrong* answer are the highest-priority targets for prompt refinement or annotation review: unanimity of error across architecturally diverse models makes model-idiosyncratic explanation implausible, pointing instead to shared training priors, prompt ambiguity, or genuine ground-truth noise. Cells in which a single model is the sole outlier—either the only model correct or the only model incorrect—index model-specific capacity gaps independent of task difficulty. Targeting qualitative analysis at consensus errors before individual outliers maximises diagnostic yield per unit of human effort.

The domain and stratification analyses converge on a specific, actionable direction for prompt improvement: a blinding rule sensitive to intervention type. Models would benefit from explicit guidance distinguishing trials in which blinding is physically feasible (pharmacological interventions with matched placebos) from those in which it is not (surgical procedures, physiotherapy,

766 medical devices), with the latter receiving a conservative default that does not
767 presuppose the availability of a blinding mechanism. This single rule targets
768 D4—the domain with the most systematic and directionally consistent errors
769 in the benchmark—and has direct implications for the High Risk recall deficit
770 identified above.

771 The recent preregistered evaluation by Nyrhi et al. [9] demonstrates that
772 automated RoB assessment with frontier LLMs is an active area of investigation.
773 That study applied four models (ChatGPT o3, DeepSeek v3, Gemini Flash 2.0,
774 Grok 3) to 100 RCTs from 10 Cochrane reviews across four clinical specialties,
775 using multimodal PDF input, and reported interobserver κ of 0.27–0.39 for RoB 1.0.
776 The present study and Nyrhi et al. share no overlapping models and differ sub-
777 stantially in corpus composition (127 vs. 10 Cochrane reviews; 24 vs. 4 clinical
778 specialties), corpus size (227 vs. 100 RCTs), and input modality (text-only pipeline
779 vs. direct PDF upload). Direct numerical comparison of performance figures is
780 therefore not warranted. Qualitatively, the κ values observed here ($\kappa = 0.325$
781 for the best model; lower for the remaining seven) fall within and below the
782 range they reported, which is consistent with expectations given the broader
783 and more heterogeneous corpus used here—more diverse corpora generally
784 present harder classification problems—and the use of a text-only rather than
785 multimodal pipeline. Both studies independently reach the same qualitative
786 conclusion: contemporary frontier LLMs do not meet the reliability threshold for
787 autonomous RoB assessment under a zero-shot, unassisted design.

788 The present study contributes three elements that go beyond that aggregate
789 conclusion. First, evaluating eight models spanning five provider families on a
790 corpus of 227 studies across 24 clinical specialties reveals that aggregate per-
791 formance is approximately stable across frontier models, shifting the central
792 question from *which model* to *which domains and intervention types* support
793 reliable assistance—a question that aggregate summaries cannot answer. Second,
794 domain-stratified analysis identifies structured heterogeneity: D1 and D2 may

795 represent nearer-term targets for assisted screening, whereas the systematic
796 leniency bias in D4 and the idiosyncratic divergence in D7 identify specific failure
797 modes that require targeted prompt improvements before reliable assistance
798 is achievable. Third, explicit degenerate baselines and the juxtaposition of Co-
799 hen’s κ with Gwet’s AC1 expose analytical pitfalls specific to this task: the margin
800 above trivially achievable strategies is narrowest for the clinically consequential
801 classes (High Risk and Unclear Risk), and the two agreement coefficients diverge
802 most sharply for the model with the highest accuracy (Claude Haiku 4.5: $\kappa = 0.325$
803 vs. $AC1 = 0.485$), cautioning against treating either coefficient in isolation when
804 class distributions are skewed.

805 Several limitations should be noted. First, this study was not pre-registered;
806 the analysis plan was developed alongside the benchmark construction, and
807 the choice of metrics, stratification variables, and comparisons was informed
808 by exploratory review of the data. Findings should therefore be interpreted as
809 exploratory and hypothesis-generating throughout, notwithstanding the use
810 of formal confidence intervals to characterise estimation uncertainty; they are
811 not the outcome of pre-specified confirmatory hypothesis tests. Second, the
812 ground-truth annotations derive from published Cochrane review reports and
813 were not independently re-annotated for this study. The Cochrane annotations
814 represent the consensus output of a structured multi-reviewer process (independ-
815 ent assessment by at least two trained reviewers, with disagreements resolved
816 by discussion or arbitration), which is the reference standard used operationally
817 by the field and the most appropriate benchmark for evaluating a tool intended
818 to assist real-world systematic review workflows. Quantifying internal reliability
819 within this corpus would require re-annotating a subsample with independent
820 trained reviewers; however, agreement between Cochrane-trained reviewers
821 and non-specialist re-annotators would conflate task difficulty with expertise
822 asymmetry—the same confound that limits the interpretability of comparisons
823 between Cochrane and external reviewers [5]. The relevant population-level IRR

estimate is therefore taken from Hartling et al. [1], who measured agreement between formally trained reviewer pairs applying the same protocol: $\kappa = 0.79$ for D1 but only $\kappa = 0.24$ – 0.37 for remaining domains, falling to $\kappa = 0.05$ – 0.09 between independent consensus pairs; independent Cochrane teams show similarly variable agreement [4]. These figures indicate that the reference standard itself carries substantial noise, placing a practical ceiling on any automated system’s agreement with published annotations that is independent of model quality. The observed model accuracy range (57.6%–62.5%) therefore reflects agreement with a reference standard that is itself imperfect across most non-D1 domains: disentangling genuine model failure from annotation noise in the ground truth is not possible without independent re-annotation of the benchmark corpus. Third, each model was queried once per study using provider-default generation parameters. This reflects the intended evaluation scope: zero-shot performance as models are available to practitioners, without artificial parameter tuning. Manipulating generation temperature would introduce heterogeneous conditions across model families, and strategies such as majority voting over repeated draws represent post-hoc optimisation outside the zero-shot benchmark design. As a consequence, differences between models separated by fewer than five percentage points should not be interpreted as definitive rank orderings in the absence of confidence intervals; these are discussed as directional trends. Fourth, all models were accessed in June 2026 via the OpenRouter API using the identifier strings reported in Table 2; providers may silently update the weights associated with a fixed identifier, so exact reproduction requires accessing the same model strings within the same inference window. Results should therefore be interpreted as reflecting the checkpoint state active in June 2026, not any future revision of the same identifier. Raw model outputs (per-cell predictions and justifications) are not publicly deposited: the full-text PDFs used as input are not all open-access and re-distribution could infringe publisher copyright; additionally, the underlying study corpus is currently under independent expert

853 medical review for a forthcoming benchmark publication, and its public release
854 is deferred to that work. Fifth, the single-agent zero-shot design examines one
855 specific slice of the capability space; few-shot prompting, retrieval augmentation,
856 fine-tuning, and multi-agent architectures were not evaluated and may
857 yield substantially different results. Sixth, training-data contamination cannot
858 be excluded, but several structural features of the evaluation design limit its
859 plausibility as a primary driver of results. Memorisation-based performance
860 would require a model to (i) have seen the particular one of the 127 distinct
861 Cochrane 2024 reviews from which each study was drawn, (ii) correctly identify
862 that the input Markdown document corresponds to a specific entry in that
863 review’s RoB table, and (iii) reproduce the reviewer’s label rather than reason
864 from the text. The Cochrane reviews sampled here were published in 2024;
865 most included RCTs appear in paywalled journals, making full-text ingestion
866 at training time less likely for the majority of papers. Moreover, the empirical
867 pattern is inconsistent with memorisation at scale: the best individual model
868 (Claude Haiku 4.5) achieved a perfect 7/7 score on only 12 of 227 studies (5.3%),
869 and even across the full ensemble of eight models all seven domains were correct
870 in only 40 studies (17.6%). Both figures are consistent with the rates expected
871 under independent domain judgements at the observed per-domain accuracy
872 ($0.60^7 \approx 2.8\%$; inflated modestly by the low intraclass correlation of $\hat{\rho} = 0.044$ to
873 roughly 4–5% per model; see Supporting Information S7 for derivation), leaving
874 no systematic excess attributable to retrieval. Targeted contamination probes
875 (e.g. membership inference or verbatim reproduction tests) were not conducted,
876 as the opaque training corpora of the evaluated models preclude rigorous analysis;
877 this remains a caveat for interpretation. Seventh, the Cochrane corpus
878 used here comprises better-reported trials than the broader trial literature, so
879 these performance estimates may be optimistic relative to unselected primary
880 research.

881 **Acknowledgements**

882 We gratefully acknowledge the financial support of the TELUS Digital Research
883 Hub.

884 **Data and Code Availability**

885 The analysis code is publicly available at <https://github.com/ACS-USP/systematic-reviews>.

886 **Conflicts of Interest**

887 The authors declare no conflicts of interest.

888 **References**

889 [1] Lisa Hartling, Michele P Hamm, Andrea Milne, Ben Vandermeer, P Lina San-
890 taguida, Mohammed Ansari, Alexander Tsertsvadze, Susanne Hempel, Paul
891 Shekelle, and Donna M Dryden. Testing the risk of bias tool showed low
892 reliability between individual reviewers and across consensus assessments
893 of reviewer pairs. *Journal of Clinical Epidemiology*, 66(9):973–981, 2013. doi:
894 10.1016/j.jclinepi.2012.07.005.

895 [2] Julian PT Higgins, James Thomas, Jacqueline Chandler, Miranda Cumpston,
896 Tianjing Li, Matthew J Page, and Vivian A Welch, editors. *Cochrane Handbook*
897 *for Systematic Reviews of Interventions*. John Wiley & Sons, 2nd edition, 2019.
898 doi: 10.1002/9781119536604.

899 [3] Julian PT Higgins, Douglas G Altman, and Jonathan AC Sterne. Assessing risk
900 of bias in included studies. In Julian PT Higgins and Sally Green, editors,
901 *Cochrane Handbook for Systematic Reviews of Interventions*, chapter 8. The
902 Cochrane Collaboration, 5.1.0 edition, 2011.

- 903 [4] Nadja Könsgen, Ognjen Barcot, Simone Heß, Livia Puljak, Käthe Goossen,
904 Tanja Rombey, and Dawid Pieper. Inter-review agreement of risk-of-bias
905 judgments varied in Cochrane reviews. *Journal of Clinical Epidemiology*, 120:
906 25–32, 2020. doi: 10.1016/j.jclinepi.2019.12.016.
- 907 [5] Susan Armijo-Olivo, Maria Ospina, Bruno R da Costa, Matthias Egger, Humam
908 Saltaji, Jorge Fuentes, Christine Ha, and Greta G Cummings. Poor reliability
909 between Cochrane reviewers and blinded external reviewers when applying
910 the Cochrane risk of bias tool in physical therapy trials. *PLOS ONE*, 9(5):
911 e96920, 2014. doi: 10.1371/journal.pone.0096920.
- 912 [6] Iain J Marshall, Joël Kuiper, and Byron C Wallace. RobotReviewer: evaluation
913 of a system for automatically assessing bias in clinical trials. *Journal of the*
914 *American Medical Informatics Association*, 23(1):193–201, 2016. doi: 10.1093/
915 jamia/ocv044.
- 916 [7] Judith-Lisa Lieberum, Markus Toews, Maria-Inti Metzendorf, Felix Heilmeyer,
917 Waldemar Siemens, Christian Haverkamp, Daniel Böhringer, Joerg J Meerpohl,
918 and Angelika Eisele-Metzger. Large language models for conducting system-
919 atic reviews: on the rise, but not yet ready for use—a scoping review. *Journal*
920 *of Clinical Epidemiology*, 181:111746, 2025. doi: 10.1016/j.jclinepi.2025.111746.
- 921 [8] Dmitry Scherbakov, Nina Hubig, Vinita Jansari, Alexander Bakumenko, and
922 Leslie A Lenert. The emergence of large language models as tools in literature
923 reviews: a large language model-assisted systematic review. *Journal of*
924 *the American Medical Informatics Association*, 32(6):1071–1086, 2025. doi:
925 10.1093/jamia/ocaf063.
- 926 [9] Lauri Nyrhi, Ville Ponkilainen, Juho Laaksonen, Lauri Kuikka, Lauri Paljakka,
927 Teemu Karjalainen, Ville M Mattila, and Ilari Kuitunen. Large language models
928 for risk-of-bias assessment in randomised clinical trials—a comparative
929 validation study. *eBioMedicine*, 126:106238, 2026. doi: 10.1016/j.ebiom.2026.
930 106238.

- [10] Jonathan AC Sterne, Jelena Savović, Matthew J Page, Pankaj Roy, Julian PT Higgins, Isabelle Boutron, Christopher J Cates, Hao-Yun Cheng, Matthew S Corbett, Sandra M Eldridge, Jonathan R Emberson, Miguel A Hernán, Sally Hopewell, Asbjørn Hróbjartsson, Daniela R Junqueira, Peter Jüni, Jamie J Kirkham, Toby Lasserson, Tianjing Li, Alexandra McAleenan, Barnaby C Reeves, Sasha Shepperd, Ian Shrier, Lesley A Stewart, Kate Tilling, Ian R White, Penny F Whiting, and Julian PT Higgins. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ*, 366:l4898, 2019. doi: 10.1136/bmj.l4898.
- [11] Bin Wang, Chao Xu, Xiaomeng Zhao, Linke Ouyang, Fan Wu, Zhiyuan Zhao, Rui Xu, Kaiwen Liu, Yuan Qu, Fukai Shang, Bo Zhang, Liquan Wei, Zhihao Sui, Wei Li, Botian Shi, Yu Qiao, Dahua Lin, and Conghui He. MinerU: An open-source solution for precise document content extraction. *arXiv preprint arXiv:2409.18839*, 2024. doi: 10.48550/arXiv.2409.18839.
- [12] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V Le, Ed H Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=1PL1NIMMrw>.
- [13] Kilem L Gwet. Computing inter-rater reliability and its variance in the presence of high agreement. *British Journal of Mathematical and Statistical Psychology*, 61(1):29–48, 2008. doi: 10.1348/000711006X126600.
- [14] Alvan R Feinstein and Domenic V Cicchetti. High agreement but low kappa: I. The problems of two paradoxes. *Journal of Clinical Epidemiology*, 43(6): 543–549, 1990. doi: 10.1016/0895-4356(90)90158-L.

956 **S1 Prompt Used for Single-Agent RoB Assessment**

957 The following prompt was used without modification across all eight models and
958 all 227 studies. Each study was assessed in seven sequential inference calls—one
959 per RoB domain—with the full Markdown text of the study report included in
960 every call as the user message, followed by the domain identifier (e.g. Domain:
961 D1). The prompt text below was supplied as the system message.

962 Goal: Systematically assess the risk of bias in a research article using
963 Cochrane RoB 1.0. Evaluate bias domain, assign a judgment (Low, High,
964 or Unclear Risk), and justify strictly from the article evidence.

965

966 ---

967

968 Preliminary Considerations

969

970 Before assessing, identify:

971 1. Intervention Type: Pharmacological | Surgical/Procedural |

972 Non-pharmacological | Device

973 2. Primary Outcome Type: Objective (mortality, lab values) |

974 Subjective (pain, QoL)

975 3. Comparator: Placebo | Active comparator | Usual care | No intervention

976

977 Critical Context: Blinding expectations vary by intervention type.

978 For pharmacological interventions, lack of placebo is a serious flaw.

979 For surgical/non-pharmacological interventions where blinding is impossible,

980 focus on compensatory measures: blinding of outcome assessors and objective

981 endpoints.

982

983 ---

984

985 Bias Domains

986

987 1. Random Sequence Generation (Selection Bias)

988 Signalling Question: Was a truly random method used?

989 Low Risk: Random method adequately described (computer-generated,
990 random tables, coin toss, dice, shuffled envelopes,
991 minimization with random element)

992 High Risk: Non-random method (alternation, date of birth, hospital
993 record number, clinician judgment)

994 Unclear Risk: States "randomized" but no method details provided

995

996 2. Allocation Concealment (Selection Bias)

997 Signalling Question: Was the allocation sequence concealed from those
998 enrolling participants until assignment?

999 Low Risk: Adequate concealment (central/telephone randomization,
1000 sequentially numbered opaque sealed envelopes, numbered
1001 drug containers, independent pharmacy)

1002 High Risk: Allocation not concealed or predictable (open list,
1003 non-opaque envelopes, alternation)

1004 Unclear Risk: Insufficient information to judge

1005 Note: Allocation concealment (before randomization) is different from
1006 blinding (after randomization).

1007

1008 3. Blinding of Participants and Personnel (Performance Bias)

1009 Signalling Questions: Were participants/personnel aware of assignments?

1010 If yes, could this affect outcomes?

1011 Pharmacological: Double-blinding expected; no placebo implies higher
1012 risk for subjective outcomes.

1013 Surgical/Non-pharmacological: Blinding may be impossible and is not
1014 automatically High Risk; check compensatory safeguards.

1015 Objective outcomes: Less susceptible to performance bias.

1016 Low Risk: Adequate blinding, OR blinding impossible with compensatory

1017 measures, OR outcome is objective/hard endpoint

1018 High Risk: No blinding + subjective outcome + no compensatory measures

1019 Unclear Risk: Insufficient blinding information

1020

1021 4. Blinding of Outcome Assessors (Detection Bias)

1022 Signalling Questions: Were assessors blinded? Is outcome objective or

1023 subjective?

1024 Low Risk: Assessors blinded, OR objective outcome unlikely to be

1025 influenced by assignment knowledge

1026 High Risk: Assessors not blinded + subjective outcome requiring

1027 interpretation

1028 Unclear Risk: Insufficient information

1029 Note: For patient-reported outcomes, participant acts as assessor.

1030

1031 5. Incomplete Outcome Data (Attrition Bias)

1032 Signalling Questions: Are data available for most participants? Are

1033 missing-data reasons balanced/unrelated to outcome?

1034 Low Risk: Minimal missingness (<5%), OR balanced missingness with

1035 unrelated reasons, OR valid ITT/imputation

1036 High Risk: High missingness (>10-20%), OR imbalance between groups,

1037 OR outcome-related missingness, OR per-protocol exclusions

1038 Unclear Risk: Missingness not adequately reported

1039

1040 6. Selective Reporting (Reporting Bias)

1041 Signalling Questions: Is protocol/registry available? Were pre-specified

1042 outcomes reported?

1043 Low Risk: All pre-specified outcomes reported, OR protocol/registry

1044 aligns with publication

1045 High Risk: Selective non-reporting (methods outcomes missing in

1046 results, changed primary outcome, significance-driven

1047 reporting)

1048 Unclear Risk: No protocol/registry and insufficient information

1049

1050 7. Other Bias

1051 Signalling Questions: Are there other bias sources (early stopping,
1052 baseline imbalance, conflicts, contamination)?

1053 Low Risk: No relevant additional bias detected

1054 High Risk: Important additional bias present

1055 Unclear Risk: Insufficient information

1056

1057 ---

1058

1059 Key Principles

1060 1. Evidence-based only: Base judgments on explicit information. Do not
1061 assume. If evidence is missing, choose "Unclear Risk".

1062 2. Cite evidence: Reference specific text or note absence of information.

1063 3. Context matters: Apply blinding expectations appropriate to intervention
1064 type.

1065 4. Impossibility vs Negligence: Not blinding when impossible differs from
1066 not blinding when feasible.

1067 5. Outcome sensitivity: Subjective outcomes require stricter blinding than
1068 objective ones.

1069

1070 ---

1071

1072 Return ONLY valid JSON with:

1073 {

1074 "judgement": "Low Risk" | "High Risk" | "Unclear Risk",

1075 "justification": "short justification"

1076 }

Table S1: Overall performance on RoB 1.0 assessment ($n = 227$ studies; 1,589 cells). Models ranked by accuracy. F1-LR, F1-HR, F1-UR: F1 for Low Risk, High Risk, and Unclear Risk computed globally across all study-domain cells. Cohen’s κ and Gwet’s AC1 are both chance-corrected agreement coefficients; they differ in how the expected chance agreement p_e is computed (see Methods). Best value in each column shown in bold. 95% confidence intervals (study-clustered bootstrap, 5,000 iterations, resampling 227 studies with replacement) are shown in brackets below each point estimate; all analyses are exploratory (see Methods, Statistical Inference).

Model	Accuracy (%)	Macro-F1	F1-LR	F1-HR	F1-UR	Cohen’s κ	Gwet’s AC1
Claude Haiku 4.5	62.5 [59.8, 65.2]	0.541 [0.511, 0.569]	0.743 [0.718, 0.767]	0.469 [0.412, 0.521]	0.413 [0.361, 0.463]	0.325 [0.286, 0.365]	0.485 [0.443, 0.526]
GLM-5	61.1 [58.7, 63.5]	0.557 [0.529, 0.582]	0.709 [0.682, 0.735]	0.441 [0.385, 0.494]	0.520 [0.479, 0.561]	0.341 [0.301, 0.380]	0.449 [0.411, 0.485]
Qwen 3.7-Plus	60.9 [58.3, 63.4]	0.525 [0.496, 0.552]	0.718 [0.691, 0.743]	0.356 [0.293, 0.413]	0.503 [0.459, 0.543]	0.306 [0.266, 0.345]	0.457 [0.417, 0.495]
Gemini 2.5 Flash	60.7 [58.2, 63.1]	0.558 [0.533, 0.582]	0.718 [0.690, 0.744]	0.474 [0.424, 0.516]	0.484 [0.443, 0.526]	0.349 [0.312, 0.384]	0.437 [0.399, 0.475]
DeepSeek v4-Flash	60.3 [57.7, 62.9]	0.558 [0.531, 0.585]	0.704 [0.676, 0.730]	0.480 [0.429, 0.530]	0.489 [0.447, 0.530]	0.337 [0.296, 0.377]	0.433 [0.394, 0.472]
Grok 4.3	60.1 [57.8, 62.5]	0.549 [0.523, 0.574]	0.696 [0.670, 0.722]	0.432 [0.376, 0.486]	0.518 [0.478, 0.556]	0.326 [0.288, 0.362]	0.433 [0.397, 0.471]
GPT-5.4	57.7 [55.1, 60.3]	0.546 [0.519, 0.572]	0.668 [0.636, 0.698]	0.464 [0.410, 0.511]	0.506 [0.468, 0.543]	0.322 [0.282, 0.361]	0.387 [0.349, 0.426]
Gemini 2.5 Pro	57.6 [55.1, 60.0]	0.543 [0.517, 0.569]	0.674 [0.643, 0.702]	0.455 [0.407, 0.501]	0.502 [0.458, 0.542]	0.331 [0.293, 0.367]	0.383 [0.344, 0.420]

1077 **S2 Overall Model Performance**

1078 **S3 Per-Domain Performance by Model**

1079 Tables S2 and S3 report accuracy (%) and Macro-F1 for each model in each of the
1080 seven RoB domains.

1081 **S4 Stratification by Intervention Type**

1082 Figure S1 shows the distribution of intervention types within each clinical domain
1083 in the 227-study corpus. Because each study can be tagged with multiple clinical
1084 domains, the counts across domains sum to more than 227; a study tagged
1085 with two domains contributes its intervention type to both. The distribution
1086 is markedly non-uniform: Pregnancy & childbirth and Complementary & alternative
1087 medicine are almost exclusively pharmacological, Effective practice &
1088 health systems and Mental health are predominantly non-pharmacological, and
1089 Orthopaedics & trauma contains no pharmacological studies. Table S4 reports
1090 Macro-F1 for each model within each intervention type. Figure S2 shows the same
1091 data as a heatmap.

Table S2: Per-domain accuracy (%) for all eight models ($n = 227$ cells per domain). Best value per domain in bold.

Model	D1	D2	D3	D4	D5	D6	D7
Claude Haiku 4.5	73.6	74.0	67.4	51.5	64.3	57.7	48.9
GLM-5	76.7	74.0	55.9	53.7	62.1	55.9	49.3
Qwen 3.7-Plus	77.5	74.9	49.8	52.0	59.9	56.4	55.5
Gemini 2.5 Flash	72.2	74.0	72.2	51.5	60.4	62.6	31.7
DeepSeek v4-Flash	74.0	70.0	67.0	57.7	54.6	55.9	42.7
Grok 4.3	75.8	70.0	59.5	54.6	58.6	60.8	41.4
GPT-5.4	75.8	70.0	64.3	52.0	51.5	55.9	34.4
Gemini 2.5 Pro	76.7	74.4	62.6	55.1	55.1	55.1	24.2

Table S3: Per-domain Macro-F1 for all eight models ($n = 227$ cells per domain). Best value per domain in bold.

Model	D1	D2	D3	D4	D5	D6	D7
Claude Haiku 4.5	0.572	0.618	0.527	0.425	0.394	0.387	0.269
GLM-5	0.632	0.657	0.462	0.479	0.514	0.443	0.345
Qwen 3.7-Plus	0.638	0.672	0.428	0.447	0.439	0.388	0.288
Gemini 2.5 Flash	0.549	0.663	0.626	0.411	0.412	0.438	0.280
DeepSeek v4-Flash	0.602	0.610	0.583	0.521	0.419	0.424	0.347
Grok 4.3	0.618	0.668	0.524	0.492	0.423	0.449	0.315
GPT-5.4	0.625	0.635	0.515	0.469	0.419	0.397	0.317
Gemini 2.5 Pro	0.621	0.671	0.481	0.478	0.413	0.475	0.195

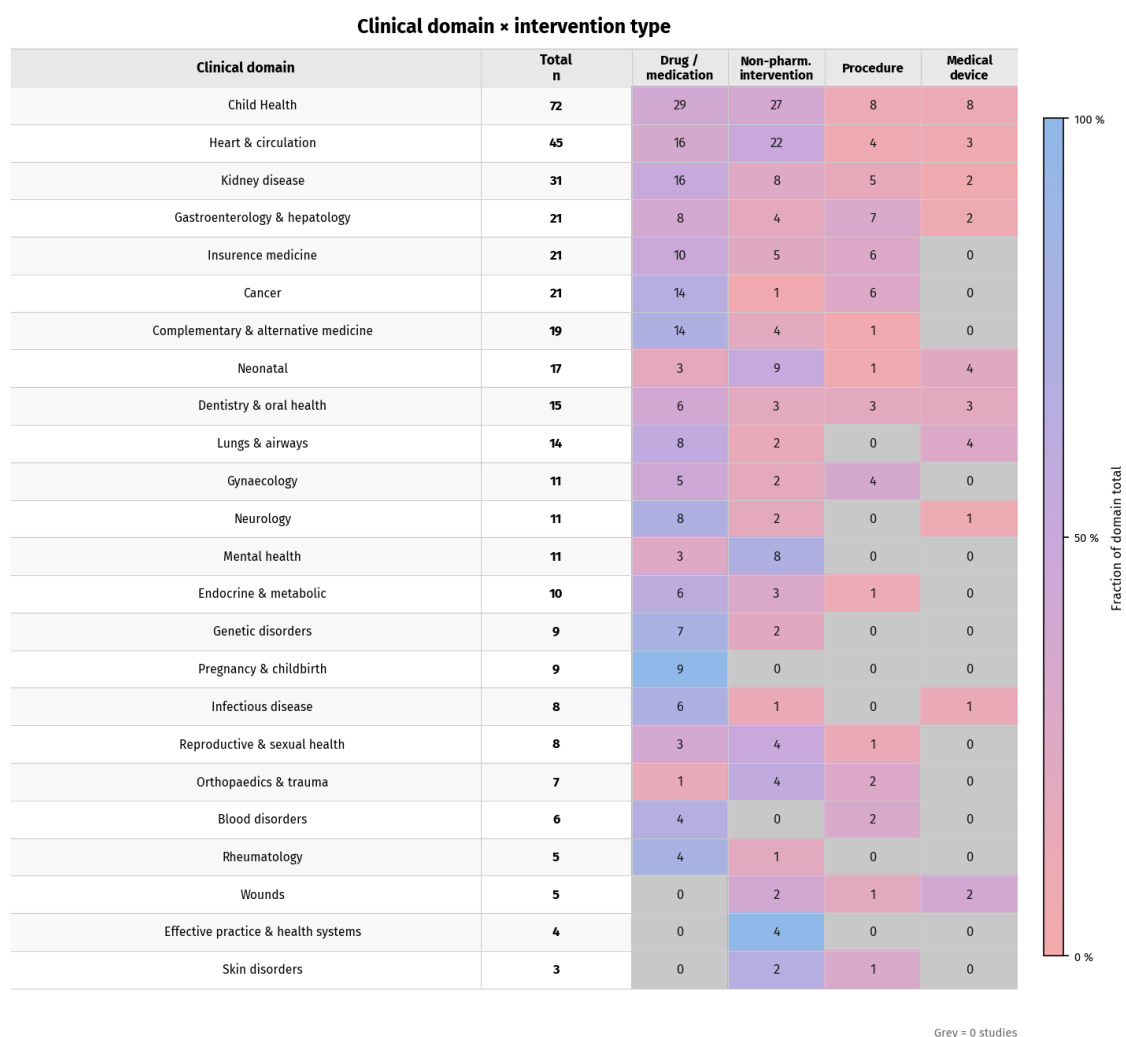


Figure S1: Distribution of intervention types across clinical domains ($n = 227$ studies; studies with multiple clinical domain tags are counted in each relevant domain). Cell colour encodes the fraction of that domain's studies belonging to each intervention type (pastel red = low proportion, pastel blue = high proportion); grey cells indicate zero studies. Domains are sorted by total study count (descending).

S5 Stratification by Clinical Domain

Table S5 reports mean Macro-F1 (averaged across all eight models) for all 24 Cochrane Review Group categories, ordered from highest to lowest mean F1. Domains with fewer than five studies are included for completeness but should be interpreted with caution.

Table S4: Macro-F1 stratified by intervention type. Numbers in parentheses are study counts (n_{studies}). Best value per row in bold. Subgroup estimates carry substantial uncertainty given the small group sizes; these analyses are descriptive only.

Model	Drug / medication ($n = 104$)	Non-pharma- cological ($n = 75$)	Procedure ($n = 30$)	Medical device ($n = 18$)
Claude Haiku 4.5	0.548	0.530	0.494	0.627
GLM-5	0.559	0.590	0.514	0.471
Qwen 3.7-Plus	0.511	0.567	0.474	0.445
Gemini 2.5 Flash	0.540	0.574	0.529	0.596
DeepSeek v4-Flash	0.548	0.582	0.520	0.543
Grok 4.3	0.552	0.561	0.519	0.531
GPT-5.4	0.515	0.595	0.530	0.502
Gemini 2.5 Pro	0.533	0.575	0.498	0.534

S6 Pairwise Model Complementarity

Table S6 reports pairwise complementarity across the eight models. Each cell (i, j) gives the percentage of the 1,589 study–domain cells in which model i produced an incorrect judgement and model j produced the correct one. High values indicate that model j recovers errors that model i misses, i.e. that j adds non-redundant information over i . The matrix is asymmetric: cell $(i, j) \neq$ cell (j, i) because the two models may have different overall accuracy.

S7 Confidence Interval Method Comparison

All confidence intervals reported in the main text were computed using study-clustered bootstrap (5,000 iterations, resampling the 227 studies with replacement). This section documents the comparison between bootstrap and two Wilson score interval variants for accuracy, and explains why the bootstrap approach was adopted for all metrics.

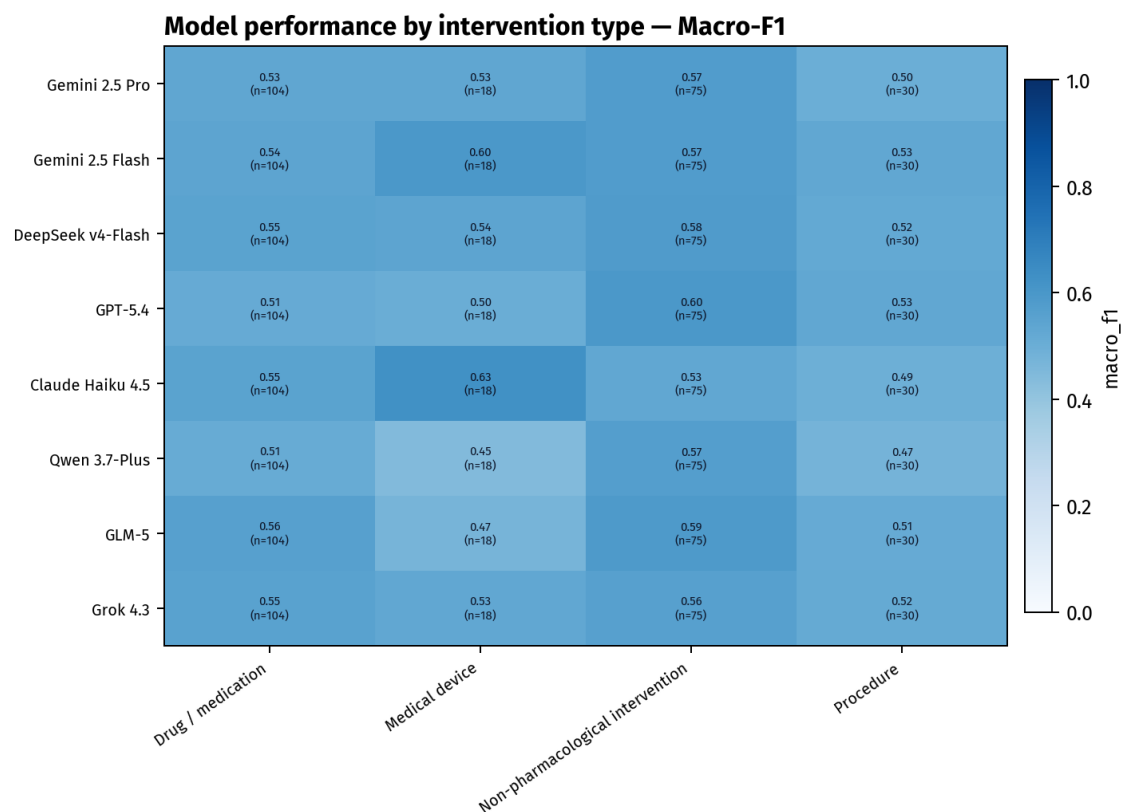


Figure S2: Macro-F1 heatmap by intervention type and model. Colour scale: 0–1.

Wilson score interval

For a proportion p observed over n independent Bernoulli trials, the Wilson 95% CI is

$$\left(\frac{np + \frac{z^2}{2}}{n + z^2} \pm \frac{z \sqrt{np(1-p) + \frac{z^2}{4}}}{n + z^2} \right), \quad (\text{S1})$$

with $z = 1.96$. Applied to accuracy ($p \approx 0.60$), two natural choices for n exist: $n = 1,589$ (all annotated cells, assuming independence) or $n = 227$ (studies as the primary sampling unit).

Intraclass correlation and effective sample size

The 1,589 cells are not independent: seven domain judgements are drawn from the same study. Pearson's one-way intraclass correlation coefficient (ICC) for the correct/wrong binary indicator across the seven domains within each study was estimated at $\hat{\rho} = 0.044$ (computed for Claude Haiku 4.5 as a representative

Table S5: Mean Macro-F1 across all eight models, stratified by clinical domain (24 Cochrane Review Groups, multi-label; study counts sum to more than 227). Ordered by descending mean F1. Estimates for smaller groups carry substantial uncertainty; these analyses are descriptive only.

Clinical domain	n_{studies}	Mean Macro-F1
Orthopaedics & trauma	7	0.680
Endocrine & metabolic	10	0.638
Rheumatology	5	0.637
Wounds	5	0.628
Dentistry & oral health	15	0.597
Lungs & airways	14	0.589
Pregnancy & childbirth	9	0.583
Neonatal	17	0.578
Complementary & alternative medicine	19	0.555
Child health	72	0.550
Gastroenterology & hepatology	21	0.538
Heart & circulation	45	0.536
Neurology	11	0.519
Cancer	21	0.494
Effective practice & health systems	4	0.487
Kidney disease	31	0.484
Mental health	11	0.481
Gynaecology	11	0.480
Insurance medicine	21	0.477
Blood disorders	6	0.476
Genetic disorders	9	0.473
Skin disorders	3	0.448
Reproductive & sexual health	8	0.437
Infectious disease	8	0.425

1121 model). The low ICC indicates that whether a model correctly judges one domain
1122 bears little information about its judgement on another domain of the same
1123 study, which is consistent with the domain-specific nature of each RoB criterion.

1124 Using the Kish (1965) design-effect formula, the effective sample size is

$$n_{\text{eff}} = \frac{n_{\text{cells}}}{1 + (k - 1)\hat{\rho}} = \frac{1,589}{1 + 6 \times 0.044} \approx 1,257, \quad (\text{S2})$$

1125 where $k = 7$ domains per study.

Table S6: Pairwise complementarity matrix. Cell (i, j) = % of 1,589 cells where row model i is incorrect and column model j is correct. Row model abbreviations: Haiku = Claude Haiku 4.5; GLM-5 = GLM-5; Qwen = Qwen 3.7-Plus; Flash = Gemini 2.5 Flash; DSV4 = DeepSeek v4-Flash; Grok = Grok 4.3; GPT = GPT-5.4; Pro = Gemini 2.5 Pro.

	Haiku	GLM-5	Qwen	Flash	DSV4	Grok	GPT	Pro
Haiku	—	12.5	10.6	9.9	12.3	12.5	12.6	12.4
GLM-5	13.9	—	7.4	12.3	9.6	8.9	9.5	7.6
Qwen	12.3	7.6	—	11.7	10.3	7.6	9.6	9.4
Flash	11.7	12.8	11.9	—	11.3	10.2	9.4	9.1
DSV4	14.5	10.4	10.8	11.7	—	9.6	9.5	9.9
Grok	14.9	9.9	8.4	10.8	9.8	—	9.0	8.4
GPT	17.4	12.9	12.7	12.4	12.1	11.4	—	10.1
Pro	17.3	11.1	12.6	12.2	12.6	11.0	10.3	—

1126 Numerical comparison

1127 Table S7 reports accuracy CIs for five models under three methods.

Table S7: Comparison of 95% confidence interval methods for overall accuracy. Width in percentage points (pp). Bootstrap: study-clustered, 5,000 iterations. Wilson n_{cells} : Equation (S1) with $n = 1,589$. Wilson n_{eff} : Equation (S1) with $n = 1,257$ (Kish formula). Wilson n_{studies} : Equation (S1) with $n = 227$.

Model	Accuracy	Wilson n_{cells} (width)	Wilson n_{eff} (width)	Bootstrap (width)
Claude Haiku 4.5	62.5%	[60.1, 64.8] (4.8)	[59.8, 65.1] (5.3)	[59.8, 65.2] (5.4)
Gemini 2.5 Flash	60.7%	[58.2, 63.0] (4.8)	[57.9, 63.4] (5.4)	[58.2, 63.1] (4.9)
DeepSeek v4-Flash	60.3%	[57.9, 62.7] (4.8)	[57.6, 63.0] (5.4)	[57.7, 62.9] (5.2)
GPT-5.4	57.7%	[55.3, 60.1] (4.9)	[54.9, 60.4] (5.4)	[55.1, 60.3] (5.2)
Gemini 2.5 Pro	57.6%	[55.1, 60.0] (4.9)	[54.8, 60.3] (5.4)	[55.1, 60.0] (4.9)
Wilson $n_{\text{studies}} = 227$		widths ≈ 12.5 – 12.8 pp (not shown; see text)		

1128 Wilson with n_{cells} underestimates uncertainty by $\sim 10\%$ relative to bootstrap
1129 (widths 4.8 pp vs. 5.1 pp). Wilson with n_{eff} virtually matches the bootstrap (widths
1130 5.3 pp vs. 5.1 pp); the two methods agree to within 0.1 pp for every model. Wilson
1131 with n_{studies} overestimates uncertainty by $\sim 2.5\times$ because it discards the domain-
1132 level information and treats each study as a single binary observation.

1133 Why bootstrap was adopted for all metrics

1134 The numerical equivalence between Wilson n_{eff} and bootstrap for accuracy con-
1135 firms that the low ICC ($\hat{\rho} = 0.044$) makes the independence assumption nearly
1136 tenable. Nevertheless, bootstrap was used uniformly for three reasons. First,
1137 Macro-F1 and Gwet's AC1 are composite statistics with no closed-form variance ex-
1138 pression; bootstrap applies without modification. Second, the paired-bootstrap
1139 procedure used for inference-mode Δ values (Table 6) jointly resamples both
1140 conditions, correctly propagating correlation between them; Wilson intervals
1141 cannot be directly adapted for paired differences of non-proportion statistics.
1142 Third, a single method for all reported CIs ensures internal consistency and
1143 simplifies interpretation.

1144 **S8 Inference Cost**

1145 Table S8 reports total billed inference costs for the eight models across the
1146 227-study benchmark, retrieved from the OpenRouter account history. All costs
1147 correspond to per-domain inference mode (seven sequential calls per study).
1148 The total cost of running the full eight-model benchmark was approximately
1149 \$244 USD, providing an order-of-magnitude reference for the practical feasibility
1150 of this type of evaluation.

1151 The pipeline permitted up to two retries per domain call in the event of API
1152 errors, timeouts, or invalid responses, and some studies required full re-queuing;
1153 the number of retry attempts was not recorded per model. Retry attempts were
1154 sufficient to ensure that all 1,589 study-domain cells received a valid prediction
1155 for every model: no cells were excluded from analysis due to unresolvable output
1156 failures. Billed costs therefore reflect total API consumption including failed calls,
1157 so absolute values should not be used to derive a cost-per-assessment figure for
1158 any individual model. The *ordering* of billed costs across models is nonetheless
1159 informative: because input tokens were identical across models (same corpus
1160 and prompt) and output token counts are of similar magnitude, differences in list
1161 price per token dominate over retry-rate variation. The observed cost ordering—
1162 DeepSeek v4-Flash cheapest, GPT-5.4 most expensive—is fully consistent with
1163 published list prices on OpenRouter (Table SS8), confirming that price per token
1164 is the primary driver of cost differences in this benchmark.

1165 For Gemini 2.5 Flash and DeepSeek v4-Flash, which were also evaluated in
1166 single-call mode (Section 3.8), billed single-call costs were lower than per-domain
1167 costs (\$2.05 vs. \$9.42 and \$0.54 vs. \$3.50 respectively). The direction of this
1168 difference is expected—single-call issues one API request per study versus seven—
1169 but the magnitude should not be interpreted as a precise cost ratio because
1170 retry counts were not recorded for either inference mode.

Table S8: Total billed inference costs for the eight models (per-domain mode, 227 studies), retrieved from OpenRouter account history, 7 July 2026. Costs include retries due to API errors or invalid responses; the number of retries was not recorded per model and cross-model comparisons should not be drawn from these figures. Total across all eight models: \$244.32.

Model	Total billed cost (USD)
DeepSeek v4-Flash	3.50
Gemini 2.5 Flash	9.42
Qwen 3.7-Plus	11.40
GLM-5	30.70
Claude Haiku 4.5	32.40
Grok 4.3	33.50
Gemini 2.5 Pro	54.10
GPT-5.4	69.30
Total	244.32

S9 Expanded Formulas for Cohen's κ and Gwet's AC1

Let $\mathbf{N} = (N_{ij})$ be the $K \times K$ confusion matrix ($K = 3$), with rows indexed by predicted class i and columns indexed by ground-truth class j , $N = \sum_{i,j} N_{ij}$ the total number of evaluated cells, and

$$\hat{p}_{k,\text{model}} = \frac{1}{N} \sum_{j=1}^K N_{kj}, \quad \hat{p}_{k,\text{gt}} = \frac{1}{N} \sum_{i=1}^K N_{ik}$$

the marginal proportions of class k in the model predictions and ground truth (row and column sums of $\mathbf{N}$ divided by N), as defined in the Methods.

Cohen's κ . The chance-expected agreement is

$$p_e^\kappa = \sum_{k=1}^K \hat{p}_{k,\text{model}} \cdot \hat{p}_{k,\text{gt}},$$

where each term $\hat{p}_{k,\text{model}} \cdot \hat{p}_{k,\text{gt}}$ is the probability that both the model and ground truth independently assign class k to a randomly selected cell. Substituting into $\kappa = (p_o - p_e^\kappa)/(1 - p_e^\kappa)$ with $p_o = \sum_k N_{kk}/N$ gives the equivalent form in raw

1181 counts:

$$\kappa = \frac{N \sum_{k=1}^K N_{kk} - \sum_{k=1}^K \left(\sum_{j=1}^K N_{kj} \right) \left(\sum_{i=1}^K N_{ik} \right)}{N^2 - \sum_{k=1}^K \left(\sum_{j=1}^K N_{kj} \right) \left(\sum_{i=1}^K N_{ik} \right)}.$$

1182 **Gwet's AC1.** The mean marginal proportion for class k is

$$\pi_k = \frac{\hat{p}_{k,\text{model}} + \hat{p}_{k,\text{gt}}}{2},$$

1183 and the chance-agreement term is

$$p_e = \frac{1}{K-1} \sum_{k=1}^K \pi_k (1 - \pi_k).$$

1184 **AC1** = $(p_o - p_e)/(1 - p_e)$ follows with p_o as above.

1185 The key difference between the two coefficients lies in their chance terms: p_e^κ
1186 depends on the *product* of the two marginals, making it sensitive to imbalance
1187 in either rater's distribution; p_e for AC1 depends on the *mean* marginal π_k , which
1188 is more stable when one rater (here, the ground truth) has highly skewed class
1189 proportions.

1190 **S10 Prompt Used for Figure Description (VLM Pipeline)**

1191 During Stage 2 of the document processing pipeline (Section 2.2), each figure
1192 placeholder was submitted to Gemini 2.5 Flash (google/gemini-2.5-flash) with
1193 the following fixed prompt. The system message was:

1194 You are converting a scientific figure into faithful Markdown text.

1195 Describe only what is visibly present in the figure.

1196 Do not infer unstated conclusions, do not mention OCR or the model,

1197 and do not use JSON.

1198 Return Markdown only.

1199 The user message was:

1200 Rewrite this scientific figure as Markdown so the image can be
 1201 removed from the extracted article.

1202

1203 Requirements:

- 1204 - Preserve scientific meaning.
- 1205 - Identify figure label and caption when visible or provided in context.
- 1206 - Describe chart/figure type.
- 1207 - Describe axes, units, legend, groups, markers, and error bars when visible.
- 1208 - Summarize visible trends and approximate plotted relationships without
- 1209 inventing exact values.
- 1210 - Mention unreadable or ambiguous areas explicitly.
- 1211 - Keep the output concise but detailed enough to replace the image in
- 1212 the article.

1213

1214 Use this Markdown template:

1215 **### Figure Description**

- 1216 - source_image: '...'
- 1217 - figure_label: ...
- 1218 - caption: ...
- 1219 - figure_type: ...
- 1220 - axes_and_units: ...
- 1221 - legend_and_groups: ...
- 1222 - visual_summary: ...
- 1223 - plotted_patterns: ...
- 1224 - unreadable_or_uncertain: ...

1225

1226 Nearby article context:

1227 {context}

1228

1229 Image path reference: {image_path}

1230 where {context} is the paragraph immediately surrounding the figure in the

1231 extracted Markdown, and `{image_path}` is the relative path to the page image
1232 extracted by MinerU. The prompt was used without modification for all figures
1233 across all 227 studies.